Copula-Based Mean-Reverting Processes: Theory and Evidence from Stablecoin Prices

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Abstract

We develop a copula-based framework for constructing mean-reverting Markov processes and evaluate its empirical content using high-frequency stablecoin prices. Mean reversion is formulated through the sign of the local conditional drift toward a target, rather than through stationarity alone. We first obtain exact conditional-mean and generator characterizations for absolutely continuous Markov copulas. We then construct observable mean-reverting processes by monotone transformations of Ornstein–Uhlenbeck and Cox–Ingersoll–Ross diffusions and by mixing a stationary transition copula with the independence copula. The latter construction has a Poisson-reset representation, preserves the invariant marginal and Markov semigroup, and adds a nonlocal reversion term to the generator. Using 524,610 exact five-minute USDT/USD transitions from 2021–2025, we find strong finite-horizon mean reversion but substantial departure from a single Gaussian OU transition. A matched boundary bootstrap rejects the OU restriction in favor of the mixed-OU density in sample. The reset component improves out-of-sample copula log scores for Gaussian and two-scale Gaussian dependence, but not uniformly across copula families, and a weakly identified Markov-switching OU is the strongest five-minute forecasting benchmark. These results support the copula separation of marginal behavior and temporal dependence while showing that structural validity, dependence fit, and predictive ranking are distinct empirical criteria.

Keywords: mean reversion; Markov copula; stablecoin; transition density; Poisson reset; out-of-sample prediction

JEL classification: C22; C58; G12; G15

1 Introduction

Mean reversion is a central organizing principle in continuous-time finance. The Gaussian specification of Vasicek (1977) and the square-root diffusion of Cox et al. (1985) remain canonical because they combine an economically interpretable restoring force with tractable transition laws and, in the interest-rate setting, tractable pricing implications. Subsequent empirical work shows, however, that the conditional distribution of financial state variables is often more nonlinear and state dependent than either model allows. Chan et al. (1992) compare alternative short-rate diffusions, Aït-Sahalia (1996) documents nonlinear drift and volatility, and Bibby and Sørensen (1995), Aït-Sahalia (2002), and Kessler (1997) develop methods for discretely observed diffusions. Standard treatments of the underlying stochastic calculus and term-structure models include Shreve (2004) and Brigo and Mercurio (2006); the CIR boundary and solution structure are further discussed by Platen and Heath (2006). These results motivate a model class that retains a precise notion of reversion while relaxing the tight link between a diffusion's marginal law and its serial dependence.

Copulas provide a natural separation. Darsow et al. (1992) relate copulas to Markov transition operators through the star product, while Chen and Fan (2006) and Chen et al. (2009) develop semiparametric estimation for stationary copula-based Markov models. The diffusion construction of Bibbona et al. (2016) further shows how monotone space–time transformations preserve serial dependence while changing the marginals, and Fang et al. (2020) studies copula-based Markov processes directly. This literature establishes the probabilistic and econometric foundations for modeling invariant marginals and temporal dependence separately. It does not, by itself, guarantee the directional restoring force required by an economic notion of mean reversion. Our theory makes that restriction explicit and identifies when it is preserved by transformations and copula mixtures.

Stablecoins provide a particularly sharp application; Ante et al. (2023) survey the rapidly growing empirical literature. Their market value is designed to remain close to a reference asset, so the relevant state variable is the deviation from the peg rather than an unrestricted cryptocurrency return. Nevertheless, deviations are neither Gaussian nor uniformly short lived. Lyons and Viswanath-Natraj (2023) connect Tether's peg stability to the design and accessibility of arbitrage, whereas Duan and Urquhart (2023) find that the speed at which deviations are corrected varies substantially across stablecoins and over time. Time variation and nonlinear stabilization also appear in Djogbenou et al. (2023) and Hui et al. (2025); Bergault et al. (2024) instead represent pegged-asset dynamics with nested OU factors. Ma et al. (2023) emphasizes the link between run incentives and the centralization of arbitrage. Tail events raise a separate run-risk dimension: Liu et al. (2023) study the Terra–Luna collapse, Ahmed et al. (2025) show how information about reserve quality can alter run incentives, and Eichengreen et al. (2025) construct market-based measures of stablecoin devaluation risk. A credible empirical model must therefore represent both ordinary correction toward the peg and occasional transitions that are too wide, persistent, or asymmetric for a single Gaussian diffusion.

This paper develops such a model in three steps. Section 2 gives a measure-theoretically precise Markov–copula foundation and distinguishes local drift reversion from finite-horizon conditional-mean reversion and stationarity. Section 3 constructs mean-reverting processes by monotone transformations of OU and CIR diffusions. The observable transition density follows from the inverse map and its Jacobian, which also reveals when a formal differential-equation solution fails as an empirical model because it is not globally monotone or does not cover the observed support. Section 4 introduces the central copula-based construction. Mixing a stationary transition copula with the independence copula under a multiplicative survival weight yields a genuine Markov semigroup, an exact transition density, and a Poisson-reset interpretation. The reset intensity augments the base mean-reversion rate without altering the invariant marginal.

The empirical analysis evaluates these restrictions without selecting models by complexity. All theoretical families with a finite-dimensional observable density are estimated or assigned a retained monotonicity, support, endpoint, or identification diagnosis. This is particularly important at five-minute frequency, where measurement effects can distort OU estimates (Holý and Tomanová, 2018). Models are compared on a common log-price conditional-density measure using an untouched 2024 validation sample, with 2025 retained as post-selection evidence. This design matters for the main conclusion. The mixed-OU kernel produces a large in-sample likelihood gain and the independent-reset component improves the validation score of Gaussian and two-scale Gaussian copulas. Yet it is redundant for the fitted Student-t copula, deteriorates the fitted Archimedean mixtures, and does not by itself reproduce the slow dependence observed across horizons. A two-state OU obtains the best five-minute predictive score, although its slow regime is weakly identified. Thus the copula-reset model is the paper's principal structural contribution, while predictive model selection remains an empirical result rather than a maintained claim.

The contribution is consequently both constructive and diagnostic. On the constructive side, the paper supplies verifiable conditions under which a copula-based process has an observable Markov density and a restoring drift, together with exact OU, CIR, reset, and pricing formulas. On the diagnostic side, it shows which formally derived transformations do not define usable one-dimensional models and which empirical extensions require an augmented state. This distinction aligns the theoretical claims with the evidence and prevents a better in-sample fit from being interpreted as universal out-of-sample or economic identification.

2 Mean-Reverting Markov Processes

2.1 Probability space, supports, and notation

Work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions. Conditional laws given a point are always versions of regular conditional probability kernels in the sense of Kallenberg (2021, Chapter 8). Such kernels are probabilistically unique only up to null sets in the conditioning variable. Equalities involving an unspecified version are therefore asserted for the relevant marginal law almost everywhere; an explicitly selected pointwise version, or continuity of both sides, is needed before an equality is asserted at every interior state.

For a distribution function F, its generalized quantile is

$$Q(u) = F^{-1}(u) := \inf\{x \in \mathbb{R} : F(x) \geq u\}, \qquad 0 < u < 1. \tag{1}$$

Quantiles are not assumed finite or differentiable at $u = 0, 1$. The support of the law at time t is denoted by $E_t = \operatorname{supp}(F_t)$. A statement that a CDF is strictly increasing means strictly increasing on the interior of its support, not on all of $\mathbb{R}$. The symbol m denotes the target mean, $\bar{r}$ a base diffusion's long-run level, and s a Gamma scale. These objects were all denoted by θ in the original manuscript and must not be conflated.

2.2 Markov–copula foundations

2.2.1 Absolutely continuous Markov copulas

Let $I = (0, 1)$ and let Leb be Lebesgue probability measure on I. Set $\Delta = \{(s, t) : 0 \leq s < t\}$. We use the standard copula and star-product terminology of Nelsen (2006) and Darsow et al. (1992), subject to the pointwise-kernel qualifications below.

Assumption 2.1 (Absolutely continuous Markov–copula density evolution)**.** *There is a* selected transition-density version*, namely a jointly Borel-measurable function $c : \Delta \times I^2 \to [0, \infty)$; write $c_{s,t}(u, v) = c(s, t, u, v)$. For every $s < t$,*

$$\int_I c_{s,t}(u, v)\,\mathrm{d}v = 1, \qquad \textit{for every } u \in I,$$

$$\int_I c_{s,t}(u, v)\,\mathrm{d}u = 1, \qquad \textit{for } \mathrm{Leb}\textit{-a.e. } v \in I, \tag{2}$$

$$c_{s,t}(u, v) = \int_I c_{s,r}(u, w) c_{r,t}(w, v)\,\mathrm{d}w, \qquad \begin{array}{c} 0 \leq s < r < t, \quad u \in I, \\ \textit{for } \mathrm{Leb}\textit{-a.e. } v \in I. \end{array} \tag{3}$$

For each fixed (s, r, t, u) in (3)*, the exceptional Lebesgue-null set in v may depend on (s, r, t, u). For $B \in \mathcal{B}(I)$, define the selected kernel*

$$K_{s,t}(u, B) := \int_B c_{s,t}(u, v)\,\mathrm{d}v, \qquad u \in I, \quad s < t, \tag{4}$$

and put $K_{t,t}(u, B) = \mathbf{1}_B(u)$. Thus "for every u" below refers to this chosen pointwise kernel version; it is stronger than merely specifying a two-dimensional density up to Leb^2*-null sets. For $s < t$, define*

$$C_{s,t}(x, y) = \int_0^x \int_0^y c_{s,t}(u, v)\,\mathrm{d}v\,\mathrm{d}u, \qquad (x, y) \in [0, 1]^2. \tag{5}$$

On the diagonal set $C_{t,t}(x,y) = \min(x,y)$ and use the identity transition operator; no Lebesgue density $c_{t,t}$ is asserted.

Proposition 2.2 (Density consistency and the copula star product)**.** *Under Assumption 2.1, (5) is a copula with density $c_{s,t}$. The kernels (4) satisfy, for every $u \in I$ and every $B \in \mathcal{B}(I)$,*

$$K_{s,t}(u,B) = \int_I K_{r,t}(w,B)K_{s,r}(u,\,\mathrm{d}w), \qquad 0 \le s < r < t. \tag{6}$$

Consequently, the operators

$$(P_{s,t}g)(u) = \int_I g(v)c_{s,t}(u,v)\,\mathrm{d}v, \qquad P_{t,t} = \mathrm{I}, \tag{7}$$

form a pointwise Markov evolution preserving the uniform law. In particular, the pair-copulas obey the star-product consistency relation

$$C_{s,t} = C_{s,r} * C_{r,t}. \tag{8}$$

More explicitly, if C, D have selected densities c_C, c_D, respectively, the density-based definition used here is

$$(C * D)(x,y) = \int_I \left\{ \int_0^x c_C(a,w)\,\mathrm{d}a \right\} \left\{ \int_0^y c_D(w,b)\,\mathrm{d}b \right\} \mathrm{d}w. \tag{9}$$

This equals the familiar formula $\int_I \partial_2 C(x,w)\partial_1 D(w,y)\,\mathrm{d}w$ when the ordinary partial derivatives are interpreted as measurable almost-everywhere representatives in the integration variable w. For every $0 \le t_0 < \cdots < t_n$, the finite-dimensional density

$$f_{t_0,\ldots,t_n}(u_0,\ldots,u_n) = \prod_{j=1}^n c_{t_{j-1},t_j}(u_{j-1},u_j) \tag{10}$$

is projectively consistent. Hence these densities define a Markov process U with uniform marginals on the continuous time index. No sample-path continuity is implied by this construction alone.

Proof. Nonnegativity and the two identities in (2) give groundedness, two-increasingness, and uniform margins. For fixed (s,r,t,u), integrate (3) over B and use Tonelli's theorem to obtain (6); hence $P_{s,r}P_{r,t} = P_{s,t}$ pointwise. Another application of Tonelli gives (9) and (8). The first marginalization in (10) uses the column normalization, the last uses the row normalization, and removal of an intermediate time uses (3). Thus the family is projectively consistent. Since I is a standard Borel space, the Kolmogorov extension theorem supplies the stated process (Kallenberg, 2021, Chapter 8). The full integral details concerning derivatives and null sets are given in Appendix A.1. $\square$

Remark 2.3 (Ordinary derivatives and modified partial Dini derivatives)*.* For the selected density version, define the pointwise conditional CDF

$$H^c_{s,t}(u,y) := K_{s,t}(u,(0,y]) = \int_0^y c_{s,t}(u,v)\,\mathrm{d}v, \qquad (u,y) \in I \times [0,1]. \tag{11}$$

For each fixed y, the fundamental theorem for Lebesgue integrals gives

$$H^c_{s,t}(u,y) = \partial_1 C_{s,t}(u,y) \quad \text{for Leb-a.e. } u. \tag{12}$$

The modified partial Dini derivative $\mathfrak{D}_1 C$ of Fang et al. (2020, Definition 2.1 and Theorems 2.1–2.2) selects a CDF-valued kernel at every source state, including points where the ordinary partial derivative is absent. It and H^c are versions of the same regular conditional law and therefore agree for Leb-a.e. source state, but need not agree on a source-null set. If $u \mapsto H^c(u, y)$ is continuous at a given u—for example under an appropriate continuous, dominated density version—then the difference quotient for $C(\cdot, y)$ converges there and all three quantities agree at that u. Thus existence of a Radon–Nikodym density alone justifies (12), not an unrestricted pointwise equality. This is precisely why the present density-only framework selects (4) explicitly rather than silently identifying a density version with an everywhere-defined derivative; see also the differentiation results in Folland (1999, Chapter 3).

Remark 2.4 (Why some statements say "every u" and others "a.e. v"). The source coordinate u indexes the selected probability kernel $K_{s,t}(u, \cdot)$, so its row normalization and kernel Chapman–Kolmogorov equation are required for every u. The target coordinate v is a Radon–Nikodym density variable, and densities and their identities are intrinsically determined only Leb-a.e. in v. Integrating a density identity that holds for a.e. v gives a measure identity for every Borel set B. Conversely, equality of two absolutely continuous kernels for every B identifies their densities only a.e. in v. These are roles of the two coordinates, not properties of the letters used to name them.

Star-product consistency of copula CDFs by itself recovers the conditional composition only for a.e. source state; it does not choose a pointwise transition function. The distinction, and sufficient regularity conditions that upgrade it, are made explicit in Theorems 3.1–3.2 of Fang et al. (2020). Assumption 2.1 imposes the stronger selected-version property directly.

Remark 2.5 (Scope of the density restriction). This article deliberately restricts attention to absolutely continuous pair-copulas because every copula used in the empirical likelihood has a density. This is a simplifying scope assumption, not a claim that all Markov copulas are absolutely continuous. Consistency (3), rather than the validity of isolated bivariate copulas at selected horizons, is what supports a continuous-time Markov model.

2.2.2 Uniformization and exact conditional-expectation identities

Assumption 2.6 (Regular marginal representation)**.** *For every t, F_t is continuous and strictly increasing on E_t°, Q_t is finite on I, and $\int_I |Q_t(u)|\,\mathrm{d}u < \infty$. The process U has the transition densities in Assumption 2.1, is adapted to $(\mathcal{F}_t)$, and is Markov with respect to this filtration. Moreover,*

$$X_t = Q_t(U_t) \quad \mathbb{P}\text{-a.s.} \tag{13}$$

Theorem 2.7 (Exact quantile–density criterion)**.** *Under Assumption 2.6, for $s < t$,*

$$\mathbb{E}[X_t \mid \mathcal{F}_s] = \int_I Q_t(v) c_{s,t}(U_s, v)\,\mathrm{d}v \quad \mathbb{P}\text{-a.s.} \tag{14}$$

Consequently, the integrable process X is a martingale with respect to $(\mathcal{F}_t)$ if and only if

$$Q_s(u) = \int_I Q_t(v) c_{s,t}(u, v)\,\mathrm{d}v \quad \text{for each fixed } s < t \text{ and for } \mathrm{Leb}\text{-a.e. } u. \tag{15}$$

The exceptional set in u may depend on the time pair (s, t). If both sides of (15) have continuous versions in u, the equality extends from a.e. u to every $u \in I$.

Proof. The Markov property of U with respect to $(\mathcal{F}_t)$ and (13) give (14). Since $Q_s(U_s) = X_s$ and U_s is uniform, the martingale identity $\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s$ is equivalent to (15) up to Lebesgue-null sets. □

Proposition 2.8 (Observable transition density)**.** *If, in addition, for every t, F_t is absolutely continuous with density $p_t > 0$ on E_t°, define, for every $x \in E_s^\circ$,*

$$K^X_{s,t}(x, B) := \int_I \mathbf{1}_B(Q_t(v))\, c_{s,t}(F_s(x), v)\,\mathrm{d}v. \tag{16}$$

This selected kernel is a regular conditional-law version for X, and a corresponding transition-density version is

$$p^X_{s,t}(x,y) = c_{s,t}(F_s(x), F_t(y))\, p_t(y), \qquad x \in E^\circ_s, \quad y \in E^\circ_t, \tag{17}$$

for every selected source state x and Lebesgue-a.e. target state y. As a conditional law, (16) *is unique only for $p_s(x)\,\mathrm{d}x$-a.e. x. The exact quantifiers for its density identities are*

$$\begin{aligned}
\int_{E^\circ_t} p^X_{s,t}(x,y)\,\mathrm{d}y &= 1, && \textit{every } x \in E^\circ_s,\\
\int_{E^\circ_s} p_s(x) p^X_{s,t}(x,y)\,\mathrm{d}x &= p_t(y), && \textit{Lebesgue-a.e. } y,\\
\int_{E^\circ_r} p^X_{s,r}(x,z) p^X_{r,t}(z,y)\,\mathrm{d}z &= p^X_{s,t}(x,y), && \begin{array}{c} s<r<t, \quad \textit{every } x \in E^\circ_s,\\ \textit{Lebesgue-a.e. } y \in E^\circ_t. \end{array}
\end{aligned} \tag{18}$$

Equivalently, the Chapman–Kolmogorov identity holds for K^X at every selected x and every Borel target set.

Proof. The selected conditional law of U_t given $U_s = F_s(x)$ is $K_{s,t}(F_s(x), \cdot)$, so its pushforward by Q_t is (16). The change of variables $v = F_t(y)$ gives (17). Row normalization gives the first identity in (18); column normalization gives marginal propagation; and the substitution $w = F_r(z)$ in (3) gives its density Chapman–Kolmogorov identity. Integrating the latter in y gives the kernel identity for every Borel set. $\square$

2.2.3 Generators and the true-martingale qualification

For bounded continuous g, use the transition operators (7). The generator, martingale-problem, and localization conventions follow Ethier and Kurtz (1986, Chapter 4); the semimartingale and class–(D) facts used below follow Protter (2005, Chapters I–III).

Definition 2.9 (Right and extended generators)**.** *For a Feller evolution, the right generator at time t is*

$$\mathcal{A}_t g = \lim_{h \downarrow 0} \frac{P_{t,t+h} g - g}{h} \tag{19}$$

whenever the limit exists in the chosen Banach-space norm. An unbounded time-dependent quantile $q(t,u)$ is said to lie in the time–space extended generator domain *if a localized Dynkin formula holds with drift $(\partial_t + \mathcal{A}_t)q$ and the required integrability is finite.*

Theorem 2.10 (Generator martingale theorem)**.** *Let $q(t,u) = Q_t(u)$ lie in the time–space extended generator domain. Suppose that, for every $T < \infty$, the stopped family*

$$\{q(\tau, U_\tau) : \tau \textit{ is a stopping time with } 0 \le \tau \le T\}$$

is uniformly integrable, that is, $q(\cdot, U_\cdot)$ is of class–(D) on every finite horizon. Then $X_t = q(t, U_t)$ is a true martingale if and only if

$$(\partial_t + \mathcal{A}_t) q(t,u) = 0 \quad \textit{for } \mathrm{d}t \otimes \mathrm{Leb}(\,\mathrm{d}u)\textit{-a.e. } (t,u). \tag{20}$$

Without the class–(D) or an equivalent uniform-integrability condition, the zero-drift equation generally establishes only a local martingale.

Proof. The extended Dynkin formula makes $q(t,U_t) - q(0,U_0) - \int_0^t (\partial_s + \mathcal{A}_s) q(s,U_s)\,\mathrm{d}s$ a local martingale. Equation (20) therefore gives a local martingale; class–(D) upgrades it to a true martingale. Conversely, the finite-variation part of a special-semimartingale decomposition is unique, so a true martingale in the stated domain has zero drift almost everywhere. The measure-theoretic details, including the passage from $\mathrm{d}t \otimes \mathbb{P}$ to $\mathrm{d}t \otimes \mathrm{Leb}$, are given in Appendix A.5. $\square$

2.2.4 Gaussian and diffusion copula generators

Let Φ and ϕ be the standard Normal CDF and density.

Proposition 2.11 (Brownian and deterministic-time-change generators)**.** *For the uniformized Brownian family and $0 < s < t$, the pair-copula is Gaussian with correlation $\rho_{s,t} = \sqrt{s/t} \in (0,1)$ and density*

$$c^G_{s,t}(u,v) = \frac{\phi_{\rho_{s,t}}\big(\Phi^{-1}(u), \Phi^{-1}(v)\big)}{\phi(\Phi^{-1}(u))\,\phi(\Phi^{-1}(v))}, \tag{21}$$

where ϕ_ρ is the standard bivariate Normal density with correlation ρ. The raw standardized Brownian example is defined on $(0,\infty)$. For $t > 0$, its local generator on $C^2_c(I)$ is

$$\mathcal{A}_t g(u) = -\frac{z\phi(z)}{t} g'(u) + \frac{\phi(z)^2}{2t} g''(u), \qquad z = \Phi^{-1}(u). \tag{22}$$

If Brownian time is replaced by a differentiable, strictly increasing h with $h(t) > 0$, replace $\rho_{s,t}$ in (21) by $\sqrt{h(s)/h(t)} \in (0,1)$, and

$$\mathcal{A}^h_t g(u) = \frac{h'(t)}{h(t)} \left[-z\phi(z) g'(u) + \frac{1}{2}\phi(z)^2 g''(u) \right]. \tag{23}$$

In particular, $h(t) = e^{2at}$ with $a > 0$ gives

$$\mathcal{A} g(u) = a\phi(z)^2 g''(u) - 2az\phi(z) g'(u). \tag{24}$$

For both clocks, $\rho_{s,t} = \rho_{s,r}\rho_{r,t}$ for $s < r < t$; Gaussian conditioning therefore gives the density Chapman–Kolmogorov identity (3). A specification whose time index includes $t = 0$ must use a clock with $h(0) > 0$ rather than $B_t/\sqrt{t}$ at the origin.

Proof. For $Z_t = B_t/\sqrt{t}$, Itô's formula gives $\mathrm{d}Z_t = -Z_t(2t)^{-1}\,\mathrm{d}t + t^{-1/2}\,\mathrm{d}B_t$. Applying Itô again to $U_t = \Phi(Z_t)$ yields (22). The deterministic time change multiplies the local clock by $h'(t)$ and evaluates the Brownian coefficient at $h(t)$, which gives (23). □

Proposition 2.12 (Uniformized diffusion generator)**.** *Let*

$$\mathrm{d}R_t = b(t, R_t)\,\mathrm{d}t + \sqrt{a(t, R_t)}\,\mathrm{d}W_t \tag{25}$$

take values in an interval (ℓ, r). Suppose that its CDF $F \in C^{1,2}$ in the interior, $p = \partial_x F > 0$, and a is locally C^1 in x. Write $F^R_t(x) = F(t,x)$ and $Q^R_t = (F^R_t)^{-1}$. Assume that, for every $s < t$, selected kernels

$$K^R_{s,t}(x, B) = \int_B p^R_{s,t}(x,y)\,\mathrm{d}y \tag{26}$$

are jointly Borel in (s,t,x), form a Markov evolution for every $x \in (\ell, r)$ and Borel B, and give the entire regular conditional law, with no additional boundary atom. Then a selected version of its pair-copula density is

$$c_{s,t}(u,v) = \frac{p^R_{s,t}\big(Q^R_s(u), Q^R_t(v)\big)}{p\big(t, Q^R_t(v)\big)}. \tag{27}$$

for $(u,v) \in I^2$. As a Radon–Nikodym derivative it is unique only Leb^2*-a.e.; the displayed formula specifies the version used by the transition kernel.*

Assume additionally that the marginal density and coefficients have enough interior regularity for the classical forward equation $p_t = -\partial_x(bp) + \frac{1}{2}\partial_{xx}(ap)$, and that the left-boundary probability flux satisfies

$$b(t,x)p(t,x) - \tfrac{1}{2}\partial_x\{a(t,x)p(t,x)\} \longrightarrow 0 \quad \text{as } x \downarrow \ell \tag{28}$$

vanishes, then the local generator of $U_t = F_t(R_t)$ is

$$\mathcal{A}_t g(u) = \left[a\,p_x + \tfrac{1}{2}a_x p\right]_{x = Q^R_t(u)} g'(u) + \frac{1}{2}\left[ap^2\right]_{x = Q^R_t(u)} g''(u). \tag{29}$$

Proof. The conditional change of variables $y = Q_t^R(v)$ gives (27). Its row normalization is the normalization of the selected kernel (26); its column normalization holds for a.e. target rank and follows from marginal propagation $p(t, y) = \int_\ell^r p(s, x)p_{s,t}^R(x, y)\,\mathrm{d}x$. With $z = Q_r^R(w)$ and $\mathrm{d}w = p(r, z)\,\mathrm{d}z$, the physical transition-density Chapman–Kolmogorov identity gives (3) for every selected source rank and a.e. target rank. Apply the time-dependent Itô formula to $F_t^R(R_t)$ and integrate the forward equation from ℓ to x. Condition (28) removes the boundary constant. A final Itô application to $g(U_t)$ gives (29). The Itô and diffusion-forward-equation results used here are standard; see Karatzas and Shreve (1991, Chapters 5–6). □

Remark 2.13. The preceding interior result covers OU directly. CIR is degenerate at zero and must be treated on $(0, \infty)$ with its nonnegative solution, interior density, and boundary behavior stated separately; a uniformly elliptic whole-line assumption cannot be invoked for CIR.

2.3 Notions and characterizations of mean reversion

2.3.1 Local, conditional, and auxiliary notions

Definition 2.14 (Forward local drift). *For an integrable real-valued Markov process with transition kernel $P_{s,t}(x, \mathrm{d}y)$, its forward local drift is*

$$b_X(t, x) = \lim_{h \downarrow 0} \frac{\int y P_{t,t+h}(x, \mathrm{d}y) - x}{h}, \tag{30}$$

whenever a finite limit exists. The definition is made for a chosen regular kernel version and for F_t-almost every $x \in E_t$; a continuous version may be extended to all interior states.

Definition 2.15 (One-dimensional Markov mean reversion). *An integrable real-valued Markov process X is* locally mean reverting toward *m if*

(i) $\lim_{t\to\infty} \mathbb{E}[X_t] = m$*;*

(ii) the forward drift (30) *exists and*

$$(x - m)b_X(t, x) < 0 \quad (x \neq m), \qquad b_X(t, m) = 0 \tag{31}$$

whenever the indicated states belong to the support.

The quantity b_X is a drift, not a "reversion rate."

Definition 2.16 (Strong conditional-mean reversion). *An integrable adapted process X has* strong conditional-mean reversion *toward m if a deterministic function $\mathcal{R}_{s,t} > 0$ exists such that*

$$\mathbb{E}[X_t - m \mid \mathcal{F}_s] = \mathcal{R}_{s,t}(X_s - m), \qquad 0 \le s < t, \tag{32}$$

$$\mathcal{R}_{s,r}\mathcal{R}_{r,t} = \mathcal{R}_{s,t}, \qquad \lim_{t\to\infty} \mathcal{R}_{s,t} = 0. \tag{33}$$

A principal case is $\mathcal{R}_{s,t} = \exp\{\int_s^t \gamma(u)\,\mathrm{d}u\}$ with continuous $\gamma(u) < 0$.

Definition 2.17 (Finite-horizon and directional classes). *For model diagnostics we use two weaker labels, neither of which is silently called global mean reversion.*

(i) A finite-horizon conditional-reversion model *satisfies* (32) *and is a one-dimensional Markov model only on a stated window $0 \le t \le T$.*

(ii) A directional-reversion model *has a drift pointing toward a threshold c, but c has not been proved equal to the limiting mean.*

An observed projection of a Markov vector is called a latent-state extension*; it is not asserted to be a one-dimensional Markov process.*

Proposition 2.18 (Strong reversion implies local reversion)**.** *Suppose Definition 2.16 holds with* $\mathcal{R}_{s,t} = \exp\{\int_s^t \gamma(u)\,\mathrm{d}u\}$*, where* γ *is continuous and negative. Then*

$$b_X(t,x) = \gamma(t)(x-m). \tag{34}$$

If X_s *is integrable, then* $\mathbb{E}[X_t] \to m$*, so* X *satisfies Definition 2.15.*

Proof. Set $s = t$ and replace t in (32) by $t+h$; differentiation at $h = 0+$ gives (34). Taking expectations in (32) and using (33) gives the limiting mean. Since $\gamma(t) < 0$, the sign condition follows. □

Remark 2.19. The converse is false in general: an instantaneous sign condition and an unconditional limiting mean do not by themselves determine every finite-horizon conditional mean. This is why the local and strong definitions are kept separate.

Proposition 2.20 (Martingale rescaling)**.** *Let* X *be an integrable Markov process satisfying Definition 2.16. Then*

$$M_t = \frac{X_t - m}{\mathcal{R}_{0,t}} \tag{35}$$

is a Markov martingale. Conversely, if M *is an integrable Markov martingale,* $\mathcal{R}_{0,t} > 0$ *is deterministic and multiplicative, and* $X_t = m + \mathcal{R}_{0,t} M_t$*, then* X *satisfies* (32)*. The two processes have the same pair-copulas. If* H_t *is the marginal CDF of* M_t*, their quantiles satisfy*

$$H_t^{-1}(u) = \frac{Q_t(u) - m}{\mathcal{R}_{0,t}}. \tag{36}$$

When the hypotheses of Theorem 2.10 hold, this is equivalent to

$$(\partial_t + \mathcal{A}_t)Q(t,u) - \gamma(t)\{Q_t(u) - m\} = 0. \tag{37}$$

Proof. Multiplicativity gives $\mathcal{R}_{0,t} = \mathcal{R}_{0,s}\mathcal{R}_{s,t}$. Hence

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = \frac{\mathcal{R}_{s,t}(X_s - m)}{\mathcal{R}_{0,t}} = M_s.$$

The converse follows by reversing this calculation. At each time the map between X_t and M_t is strictly increasing and affine, so it preserves pair-copulas and gives (36). Applying the generator martingale theorem to that quantile and differentiating $\mathcal{R}_{0,t}$ gives (37). □

2.3.2 Exact copula and generator characterizations

Theorem 2.21 (Exact copula criterion for strong mean reversion)**.** *Under Assumption 2.6,* X *satisfies* (32) *if and only if*

$$\int_I \{Q_t(v) - m\} c_{s,t}(u,v)\,\mathrm{d}v = \mathcal{R}_{s,t}\{Q_s(u) - m\} \tag{38}$$

for each fixed $s < t$ *and for* Leb*-almost every* u*; the exceptional set may depend on* (s,t)*. If both sides admit continuous versions in* u*, the equality extends to every* $u \in I$*.*

Proof. By (14), the left side of (38), evaluated at $u = U_s$, equals $\mathbb{E}[X_t - m \mid \mathcal{F}_s]$. Its right side then equals $\mathcal{R}_{s,t}(X_s - m)$. Since U_s is uniform, equality almost surely is equivalent to the displayed density identity for Leb-almost every u. □

Proposition 2.22 (Generator sign criterion)**.** *Suppose $Q(t,u)$ is in the time–space extended generator domain and the local drift limit can be interchanged with the quantile representation. Then*

$$b_X(t, Q_t(u)) = (\partial_t + \mathcal{A}_t)Q(t,u). \tag{39}$$

Consequently, after separately imposing the limiting-mean condition, Definition 2.15 is equivalent to

$$\operatorname{sgn}\{(\partial_t + \mathcal{A}_t)Q(t,u)\} = \operatorname{sgn}\{m - Q_t(u)\} \tag{40}$$

for almost every interior u, with zero drift at $Q_t(u) = m$.

Proof. Insert $X_t = Q_t(U_t)$ into Definition 2.14 and use the definition of the time–space extended generator. Strict monotonicity converts the physical-state sign condition into (40). A complete kernel-integral proof, with separate bounded and weighted conditions for the moving time difference quotient, is given in Appendix A.3. ☐

2.3.3 Stationary Gaussian copula and Gamma-margin diagnostics

Let Z be the stationary standardized OU process

$$\mathrm{d}Z_t = -aZ_t\,\mathrm{d}t + \sqrt{2a}\,\mathrm{d}W_t, \qquad a > 0, \qquad Z_t \sim \mathcal{N}(0,1). \tag{41}$$

Its copula is Gaussian with correlation $e^{-a(t-s)}$.

Proposition 2.23 (Stationary Gaussian-copula sign criterion)**.** *Let F have a finite mean m, let $g(z) = F^{-1}(\Phi(z))$, and suppose g is strictly increasing, C^2, and in the extended generator domain of* (41)*. Let z_F be the unique number satisfying $g(z_F) = m$. If*

$$\operatorname{sgn}\{g''(z) - zg'(z)\} = \operatorname{sgn}(z_F - z) \quad \textit{for every } z \in \mathbb{R}, \tag{42}$$

then $X_t = g(Z_t)$ is locally mean reverting toward m.

Proof. Itô's formula gives local drift $a\{g''(z) - zg'(z)\}$. Since g is strictly increasing, $\operatorname{sgn}(m - g(z)) = \operatorname{sgn}(z_F - z)$. Stationarity gives $\mathbb{E}[X_t] = m$ at every time. Integrability and the extended-domain assumption justify the drift calculation. ☐

Example 2.24 (Stationary Gamma margin: a necessary root is not sufficient). Let G_ν be the quantile of $\mathrm{Gamma}(\nu, 1)$, with $\nu > 0$, and set $Q(u) = sG_\nu(u)$, $s > 0$. Define

$$u_\nu = F_{\nu,1}(\nu), \qquad z_\nu = \Phi^{-1}(u_\nu). \tag{43}$$

At the target mean $m = \nu s$, the zero-drift requirement reduces to

$$\Gamma(\nu)\left(\frac{e}{\nu}\right)^{\nu}\phi(z_\nu) - 2z_\nu = 0. \tag{44}$$

Equation (44) is only necessary. The model is locally mean reverting only if the full condition

$$\operatorname{sgn}\{\mathcal{A}Q(u)\} = \operatorname{sgn}\{\nu s - Q(u)\} \quad \text{for almost every } u \in I \tag{45}$$

also holds, with the required endpoint integrability. No existence claim is made from the scalar root alone.

Example 2.25 (Time-varying Gamma scale). Let $Q_t(u) = s(t)G_\nu(u)$, where $s(t) > 0$ is differentiable and $s(t) \to s_\infty \in (0, \infty)$. Write $z = \Phi^{-1}(u)$ and

$$\mathscr{G}_\nu(u) = 2\phi(z)zG'_\nu(u) - \phi(z)^2 G''_\nu(u). \tag{46}$$

For the Gaussian-copula generator (24), the quantile drift is

$$D_t(u) = s'(t)G_\nu(u) - as(t)\mathscr{G}_\nu(u). \tag{47}$$

The correct target is $m = \nu s_\infty$, and the necessary and sufficient local sign test is

$$\operatorname{sgn} D_t(u) = \operatorname{sgn}\{m - s(t)G_\nu(u)\} \quad \text{for almost every } u \in I. \tag{48}$$

If u_t solves $s(t)G_\nu(u_t) = m$, then

$$\frac{s'(t)}{s(t)} = a\frac{\mathscr{G}_\nu(u_t)}{G_\nu(u_t)} \tag{49}$$

is a necessary zero-point equation, not a sufficient global condition. Its limit also inherits the stationary root requirement (44).

3 Mean-Reverting Processes from Monotone Diffusion Transforms

3.1 Construction theorem and observable density

Let R be a Markov diffusion on an interval E,

$$\mathrm{d}R_t = b(R_t)\,\mathrm{d}t + \sigma(R_t)\,\mathrm{d}W_t, \qquad \mathcal{L}h = bh_x + \tfrac{1}{2}\sigma^2 h_{xx}. \tag{50}$$

Theorem 3.1 (Strict diffusion-transform theorem)**.** *Let $f \in C^{1,2}([0,\infty) \times E)$ and suppose that $x \mapsto f(t,x)$ is strictly increasing for every relevant t. Let γ be continuous and let*

$$\mathcal{R}_{s,t} = \exp\left\{\int_s^t \gamma(u)\,\mathrm{d}u\right\}. \tag{51}$$

Assume that

$$(\partial_t + \mathcal{L})f(t,x) = \gamma(t)\{f(t,x) - m\} \tag{52}$$

and, for every $T < \infty$,

$$\mathbb{E}\int_0^T \mathcal{R}_{0,t}^{-2}\sigma(R_t)^2 f_x(t,R_t)^2\,\mathrm{d}t < \infty. \tag{53}$$

Then $Y_t = f(t,R_t)$ is Markov and

$$\mathbb{E}[Y_t - m \mid \mathcal{F}_s] = \mathcal{R}_{s,t}(Y_s - m). \tag{54}$$

If $\gamma(t) < 0$, $\mathcal{R}_{s,t} \to 0$, and the initial first moment is finite, then Y has strong and local mean reversion toward m.

Conversely, if the strictly monotone $C^{1,2}$ transform satisfies (54) *and its semimartingale decomposition is integrable, uniqueness of the drift part yields* (52)*.*

Proof. Itô's formula and (52) give

$$\mathrm{d}(Y_t - m) = \gamma(t)(Y_t - m)\,\mathrm{d}t + \sigma(R_t)f_x(t,R_t)\,\mathrm{d}W_t.$$

Multiplying by $\mathcal{R}_{0,t}^{-1}$ removes the drift. Condition (53) makes the resulting stochastic integral a true square-integrable martingale on every finite horizon, proving (54). Strict invertibility transfers the Markov property from R to Y. The converse follows from uniqueness of the finite-variation part. These Itô-integral and semimartingale-uniqueness steps are covered, for example, by Karatzas and Shreve (1991) and Protter (2005). $\square$

Proposition 3.2 (Transition density under a monotone transform)**.** *Suppose a selected transition kernel of R has density $p_{s,t}(x_s, x_t)$ for every source state x_s, and the observations y_j lie in $f(j, E)$ with unique inverse states $x_j = f^{-1}(j, y_j)$ for $j = s, t$. Then*

$$q_{s,t}(y_s, y_t) = p_{s,t}(x_s, x_t) \left| \frac{\partial f^{-1}(t, y_t)}{\partial y_t} \right| = \frac{p_{s,t}(x_s, x_t)}{|f_x(t, x_t)|}. \tag{55}$$

for every selected source observation y_s and Lebesgue-a.e. target observation y_t. This is a valid interior density only where the support, uniqueness, and Jacobian conditions hold. For the strictly increasing transforms used here, the probability ranks are unchanged. Hence the pair-copula CDFs satisfy the pointwise identity

$$C^Y_{s,t}(u, v) = C^R_{s,t}(u, v), \qquad (u, v) \in [0, 1]^2, \tag{56}$$

and their Radon–Nikodym densities satisfy

$$c^Y_{s,t}(u, v) = c^R_{s,t}(u, v) \quad \text{for } \mathrm{Leb}^2\text{-a.e. } (u, v). \tag{57}$$

One may, and in the empirical likelihood does, select the same pointwise density version on both sides. The copula invariance in (56) *is the standard invariance under strictly increasing marginal transformations; see Nelsen (2006, Chapter 2).*

Proof. Condition on the uniquely recovered state $R_s = x_s$ and apply the one-dimensional change-of-variables theorem at time t. Moreover, $F^Y_t(f(t, x)) = F^R_t(x)$ for every interior x. Substitution into the joint CDF proves (56); uniqueness of Radon–Nikodym derivatives proves (57). □

3.2 OU base process

Use the centered OU process

$$\mathrm{d}R_t = -\kappa R_t \, \mathrm{d}t + \sigma \, \mathrm{d}W_t, \qquad \kappa > 0, \quad \sigma > 0. \tag{58}$$

For $\Delta = t - s$,

$$R_t \mid R_s = x \sim \mathcal{N}\left(e^{-\kappa\Delta} x, \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa\Delta}) \right). \tag{59}$$

The noncentered OU is obtained by replacing R_t with $R_t - \bar{r}$. It therefore satisfies

$$\mathbb{E}[R_t - \bar{r} \mid \mathcal{F}_s] = e^{-\kappa(t-s)}(R_s - \bar{r}), \qquad \lim_{t\to\infty} \mathbb{E}[R_t \mid \mathcal{F}_s] = \bar{r}. \tag{60}$$

Set $\gamma = -d$ with $d > 0$. The transform PDE is

$$f_t + \tfrac{1}{2}\sigma^2 f_{xx} - \kappa x f_x + d(f - m) = 0. \tag{61}$$

Example 3.3 (Affine transformed OU). For $A_0 > 0$ and $B_0 \in \mathbb{R}$,

$$f(t, x) = m + A_0 e^{(\kappa - d)t} x + B_0 e^{-dt} \tag{62}$$

solves (61) and is strictly increasing on $\mathbb{R}$. If the base OU is stationary, then

$$Y_t \sim \mathcal{N}\left(m + B_0 e^{-dt}, \frac{A_0^2 \sigma^2}{2\kappa} e^{2(\kappa - d)t} \right). \tag{63}$$

The variance may grow when $\kappa > d$; this does not contradict the first-moment identity (54), but the transformed process is then not stationary.

Proof. Substitution leaves the coefficient equations $A' + (d - \kappa)A = 0$ and $B' + dB = 0$. The derivative $f_x = A_0 e^{(\kappa-d)t}$ is positive. The marginal law follows from the affine Normal transformation. $\square$

Example 3.4 (Exponential transformed OU). Let

$$\begin{aligned} f(t,x) &= m + e^{A(t)x+B(t)} + C(t), &(64)\\ A(t) &= A_0 e^{\kappa t}, &(65)\\ B(t) &= B_0 - \frac{\sigma^2 A_0^2}{4\kappa}(e^{2\kappa t} - 1) - dt, &(66)\\ C(t) &= C_0 e^{-dt}. &(67)\end{aligned}$$

If $A_0 > 0$, this is strictly increasing and its support is $(m + C(t), \infty)$. It is positive-valued when $m + C(t) \geq 0$. For stationary R, the shifted variable $Y_t - m - C(t)$ is lognormal with log mean $B(t)$ and log variance $A(t)^2\sigma^2/(2\kappa)$.

Proof. Substitution in (61) gives

$$A' = \kappa A, \qquad B' = -\tfrac{1}{2}\sigma^2 A^2 - d, \qquad C' = -dC,$$

whose solution is (65)–(67). The derivative $f_x = Ae^{Ax+B}$ is positive. Gaussian exponential moments verify (53) on finite horizons. $\square$

Proposition 3.5 (OU Kummer candidate is not an admissible monotone model)**.** *Let $J(p,q,z)$ solve $zJ_{zz} + (q-z)J_z - pJ = 0$. The separated family*

$$f(t,x) = m + e^{-\zeta t} J\left(\frac{\zeta - d}{2\kappa}, \frac{1}{2}, \frac{\kappa x^2}{\sigma^2}\right) \tag{68}$$

solves (61) *wherever it is smooth. Every nonconstant member is, however, noninjective on the OU state space $\mathbb{R}$ and therefore does not satisfy Theorem 3.1 or Proposition 3.2.*

Proof. Direct substitution gives the stated Kummer equation. The spatial dependence is through x^2, so $f(t,x) = f(t,-x)$ for every x. A nonconstant even function cannot be one-to-one on $\mathbb{R}$. $\square$

Remark 3.6. A density formed by summing two inverse branches would define a different observation model. The transformation would no longer preserve the base copula, and the observed scalar process need not be Markov; this is outside the present one-dimensional monotone framework.

3.3 CIR base process

Let

$$\mathrm{d}R_t = \kappa(\bar{r} - R_t)\,\mathrm{d}t + \sigma\sqrt{R_t}\,\mathrm{d}W_t, \qquad \kappa > 0, \quad \bar{r} > 0, \quad \sigma > 0, \tag{69}$$

with state space $[0,\infty)$. The Feller condition $2\kappa\bar{r} \geq \sigma^2$ is a convenient sufficient condition for zero to be unattainable when the initial state is positive. For $\Delta = t - s$, the exact transition law is

$$R_t \mid R_s = x \overset{d}{=} c_\Delta \chi'^2_\nu(\lambda_\Delta), \tag{70}$$

$$c_\Delta = \frac{\sigma^2(1 - e^{-\kappa\Delta})}{4\kappa}, \qquad \nu = \frac{4\kappa\bar{r}}{\sigma^2}, \qquad \lambda_\Delta = \frac{4\kappa e^{-\kappa\Delta}x}{\sigma^2(1 - e^{-\kappa\Delta})}. \tag{71}$$

The conditional mean has the same linear reversion form as OU:

$$\mathbb{E}[R_t - \bar{r} \mid \mathcal{F}_s] = e^{-\kappa(t-s)}(R_s - \bar{r}). \tag{72}$$

When started from its invariant law,

$$R_t \sim \mathrm{Gamma}\left(\frac{2\kappa\bar{r}}{\sigma^2}, \frac{\sigma^2}{2\kappa}\right), \tag{73}$$

where the second parameter is scale. With $\gamma = -d$, the corrected transform PDE is

$$f_t + \tfrac{1}{2}\sigma^2 x f_{xx} + \kappa(\bar{r} - x)f_x + d(f - m) = 0. \tag{74}$$

The boundary statement, noncentral chi-square transition law, conditional mean, and invariant Gamma law are the standard CIR results; see Cox et al. (1985).

Example 3.7 (Affine transformed CIR). For $A_0 > 0$ and $B_0 \in \mathbb{R}$, define

$$A(t) = A_0 e^{(\kappa-d)t}, \tag{75}$$

$$B(t) = m - A_0\bar{r}e^{(\kappa-d)t} + B_0 e^{-dt}, \tag{76}$$

$$f(t,x) = A(t)x + B(t). \tag{77}$$

Then f solves (74), is strictly increasing on the CIR state space, and has support $[B(t), \infty)$. The original manuscript's formula is the special case $m = \bar{r}$ after a harmless reparameterization of B_0.

Proof. The affine coefficient equations are $A' + (d - \kappa)A = 0$ and $B' + d(B - m) + \kappa\bar{r}A = 0$. Equations (75)–(76) solve them, and $f_x = A > 0$. Polynomial moments of CIR verify the finite-horizon martingale condition. □

Proposition 3.8 (Quadratic CIR transform: exact finite-horizon condition)**.** *Let*

$$f(t,x) = A(t)x^2 + B(t)x + C(t), \qquad A_0 > 0, \tag{78}$$

and set

$$\eta = \frac{A_0(\sigma^2 + 2\kappa\bar{r})}{\kappa}. \tag{79}$$

The coefficient solution is

$$A(t) = A_0 e^{(2\kappa-d)t}, \tag{80}$$

$$B(t) = e^{(\kappa-d)t}\{B_0 - \eta(e^{\kappa t} - 1)\}, \tag{81}$$

$$C(t) = m + \frac{\bar{r}\eta}{2}e^{(2\kappa-d)t} - \bar{r}(B_0 + \eta)e^{(\kappa-d)t} + \left(C_0 - m + \bar{r}B_0 + \frac{\bar{r}\eta}{2}\right)e^{-dt}. \tag{82}$$

It solves (74)*. On a fixed window* $[0,T]$*, the map* $x \mapsto f(t,x)$ *is increasing on* $[0,\infty)$ *for every* $t \le T$ *if and only if*

$$B_0 \ge \eta(e^{\kappa T} - 1). \tag{83}$$

Under a strict inequality the interior inverse Jacobian is strictly finite. No finite B_0 *makes a nondegenerate quadratic transform globally increasing for all* $t \ge 0$ *under the standard CIR parameter restrictions.*

Proof. Matching the coefficients of $x^2, x, 1$ gives

$$\begin{aligned} A' + (d - 2\kappa)A &= 0, \\ B' + (d - \kappa)B + (\sigma^2 + 2\kappa\bar{r})A &= 0, \\ C' + d(C - m) + \kappa\bar{r}B &= 0, \end{aligned}$$

which yields (80)–(82). Since $f_x = 2A(t)x + B(t)$ and $A(t) > 0$, monotonicity on the nonnegative half-line is equivalent to $B(t) \geq 0$. Formula (81) is nonnegative for all $t \leq T$ exactly under (83). Its bracket tends to $-\infty$ as $t \to \infty$, proving the global impossibility. □

Remark 3.9 (Why $\kappa < 0$ is not the same CIR model). Algebraically, when $\kappa < 0$ the quadratic map can be globally increasing if $A_0 > 0$ and $\min\{B_0, B_0 + \eta\} \geq 0$. This does not rescue the standard CIR model. If $\bar{r} > 0$, the boundary drift $\kappa\bar{r} < 0$ points out of the nonnegative state space. If $\bar{r} < 0$ is chosen to make the intercept positive, the drift is explosive and there is no invariant Gamma law or mean reversion. Such a process would be a separately defined supercritical square-root diffusion, not a relaxation of (69).

Corollary 3.10 (Quadratic inverse and density)**.** *Under the strict finite-window condition and for an interior observation $y > C(t)$, the unique nonnegative inverse is*

$$x = \frac{2\{y - C(t)\}}{B(t) + \sqrt{B(t)^2 + 4A(t)\{y - C(t)\}}}, \qquad \left|\frac{\partial x}{\partial y}\right| = \frac{1}{2A(t)x + B(t)}. \tag{84}$$

Together with (70)*, this gives the finite-horizon observation density through* (55)*.*

Example 3.11 (Exponential transformed CIR). Let

$$f(t, x) = m + e^{A(t)x + B(t)} + D(t), \tag{85}$$

$$A(t) = \frac{2\kappa}{\sigma^2 + (2\kappa/A_0 - \sigma^2)e^{-\kappa t}}, \tag{86}$$

$$\begin{aligned} B(t) &= -dt - \frac{2\kappa\bar{r}}{\sigma^2}\log\big[(2\kappa/A_0 - \sigma^2) + \sigma^2 e^{\kappa t}\big] \\ &\quad + B_0 + \frac{2\kappa\bar{r}}{\sigma^2}\log(2\kappa/A_0), \end{aligned} \tag{87}$$

$$D(t) = D_0 e^{-dt}. \tag{88}$$

For $A_0 > 0$, the logarithm arguments and denominator are positive and the map is strictly increasing. Its support is

$$[m + D(t) + e^{B(t)}, \infty). \tag{89}$$

If the base CIR is stationary, a sufficient and necessary initial exponential moment restriction is

$$0 < A_0 < \frac{2\kappa}{\sigma^2}. \tag{90}$$

Under these conditions the positive local-martingale component has constant expectation and is therefore a true martingale; this verifies the conditional mean identity even when the square-integrability criterion (53) is stronger than needed.

Proof. The coefficient equations are

$$A' + \tfrac{1}{2}\sigma^2 A^2 - \kappa A = 0, \qquad B' + d + \kappa\bar{r}A = 0, \qquad D' + dD = 0.$$

Their solution is (86)–(88). Since $R_t \geq 0$, the minimum of the exponential term is $e^{B(t)}$, proving (89). The stationary Gamma MGF is finite exactly below rate $2\kappa/\sigma^2$, which yields (90). More explicitly, put $p_0 = \sigma^2 A_0/(2\kappa) \in (0, 1)$ and $q = 2\kappa\bar{r}/\sigma^2$. Then

$$B(t) = B_0 - dt - q\log(1 - p_0 + p_0 e^{\kappa t}), \qquad 1 - \frac{\sigma^2 A(t)}{2\kappa} = \frac{1 - p_0}{1 - p_0 + p_0 e^{\kappa t}}.$$

Using the stationary Gamma MGF shows

$$\mathbb{E}\exp\{dt + A(t)R_t + B(t)\} = e^{B_0}(1-p_0)^{-q},$$

which is constant. The nonnegative local-martingale component is therefore a true martingale. □

3.4 CIR special functions and the corrected reduction

Let $M(p,q,z) = {}_1F_1(p;q;z)$ denote Kummer's function and $U(p,q,z)$ Tricomi's function, with limiting definitions used at exceptional integer parameters. We use the definitions, derivative identities, and asymptotic conventions of Olver et al. (2010, Chapter 13). General solution methods for the linear partial differential equations used in the transform calculations are collected in Polyanin and Nazaikinskii (2015).

Proposition 3.12 (CIR Kummer separated family)**.** *Set*

$$p = \frac{\zeta - d}{\kappa}, \qquad q = \frac{2\kappa\bar{r}}{\sigma^2}, \qquad z = \frac{2\kappa x}{\sigma^2}. \tag{91}$$

If

$$J(p,q,z) = C_1 M(p,q,z) + C_2 U(p,q,z) \tag{92}$$

is a well-defined branch satisfying the relevant boundary conditions, then

$$f(t,x) = m + e^{-\zeta t} J(p,q,z) \tag{93}$$

solves (74)*. It defines a model in the present framework only if the selected branch is strictly increasing on* $[0,\infty)$*, has a unique stable inverse on its image, and satisfies the moment condition in Theorem 3.1 for the stated initial law.*

For example, $p, q > 0$*,* $C_1 > 0$*,* $C_2 = 0$ *gives*

$$\partial_z M(p,q,z) = \frac{p}{q} M(p+1, q+1, z) > 0, \tag{94}$$

but this branch is not integrable under the stationary Gamma law because of its e^z *asymptotic term. It may be integrable at each finite time from a deterministic initial state. Thus the separated equation is correct, whereas an unspecified "special-function CIR model" is not a complete statistical model.*

Proof. Substitution gives $zJ_{zz} + (q-z)J_z - pJ = 0$. The derivative identity and standard positive series for M prove monotonicity in the stated branch. The cited Kummer asymptotic is $M(p,q,z) \sim \Gamma(q)e^z z^{p-q}/\Gamma(p)$; multiplying by the invariant $z^{q-1}e^{-z}$ density leaves a nonintegrable z^{p-1} tail for $p > 0$. □

Proposition 3.13 (The second CIR transform in the original manuscript is false)**.** *The exponential change of variables claimed in the original manuscript does not transform* (74) *into* $u_\tau = zu_{zz}$*; direct substitution has a nonzero residual for generic parameters. A correct direct reduction is*

$$f(t,x) - m = e^{-dt}u(\tau, z), \qquad z = xe^{\kappa t}, \qquad \tau = \frac{\sigma^2}{2\kappa}(1 - e^{\kappa t}), \tag{95}$$

which yields

$$u_\tau = zu_{zz} + \frac{2\kappa\bar{r}}{\sigma^2}u_z. \tag{96}$$

Therefore the three functions listed after the false reduction are not thereby solutions of the original CIR transform PDE.

Proof. Differentiate (95) and substitute into (74); the $\kappa z u_z$ terms cancel and the remaining terms give (96). The coefficient of u_z is strictly positive under the standard CIR parameters and cannot generally be discarded. □

4 Mean-Reverting Processes from Mixed Copula Families

4.1 Transition-density construction and preservation of mean reversion

Let $P_{s,t}$ be a Markov evolution on a state space E with common invariant probability law π; thus the kernel Chapman–Kolmogorov identity holds for every selected source state x and every measurable target set. Assume that, for $s < t$, selected jointly measurable versions are dominated by a common σ-finite measure μ_0,

$$P_{s,t}(x,\,\mathrm{d}y) = p_{s,t}(x,y)\mu_0(\,\mathrm{d}y), \qquad \pi(\,\mathrm{d}y) = p_\pi(y)\mu_0(\,\mathrm{d}y).$$

Here the first equality is imposed for every x as an equality of measures; the densities in y remain unique only μ_0-a.e. Define the independence/reset operator

$$\Pi(x,A) = \pi(A), \qquad (\Pi h)(x) = \int_E h(z)\pi(\,\mathrm{d}z). \tag{97}$$

Let $\lambda : [0,\infty) \to [0,\infty)$ be right-continuous and locally bounded, and set

$$a_{s,t} = \exp\left\{-\int_s^t \lambda(u)\,\mathrm{d}u\right\}. \tag{98}$$

Theorem 4.1 (Consistent reset density mixture)**.** *The kernels*

$$P^{\mathrm{mix}}_{s,t} = a_{s,t}P_{s,t} + (1-a_{s,t})\Pi \tag{99}$$

form a Markov evolution and preserve π. Their transition densities are

$$p^{\mathrm{mix}}_{s,t}(x,y) = a_{s,t}p_{s,t}(x,y) + (1-a_{s,t})p_\pi(y). \tag{100}$$

where the right-hand side defines the selected version; its density Chapman–Kolmogorov identity holds for every x and μ_0-a.e. y, while the corresponding kernel identity holds for every x and every measurable target set. Assume additionally that $E \subseteq \mathbb{R}$, the process is started from π, the invariant CDF F_π is continuous and strictly increasing on the interior of its support, and the stationary base pair-copula has density $c_{s,t}$. Then the mixed pair-copula and its density are

$$C^{\mathrm{mix}}_{s,t} = a_{s,t}C_{s,t} + (1-a_{s,t})\Pi_C, \qquad \Pi_C(u,v) = uv, \tag{101}$$

$$c^{\mathrm{mix}}_{s,t}(u,v) = a_{s,t}c_{s,t}(u,v) + (1-a_{s,t}). \tag{102}$$

The copula identity (101) *is pointwise on* $[0,1]^2$*. Equation* (102) *is a* Leb^2*-a.e. density identity, and its right-hand side is the selected version used by the mixed kernel; with that selection, its density consistency holds for every source rank u and a.e. target rank v. If the base generator is $\mathcal{G}_t$, then, on any invariant generator domain whose functions are integrable under π,*

$$\mathcal{G}^{\mathrm{mix}}_t h(x) = \mathcal{G}_t h(x) + \lambda(t)\left\{\int_E h(z)\pi(\,\mathrm{d}z) - h(x)\right\}. \tag{103}$$

Proof. The survival weights are multiplicative. Invariance gives $P\Pi = \Pi P = \Pi$ and $\Pi^2 = \Pi$. Expanding $P^{\mathrm{mix}}_{s,r}P^{\mathrm{mix}}_{r,t}$ therefore reduces it to (99) with weight $a_{s,r}a_{r,t} = a_{s,t}$. Taking densities gives (100); uniformization under stationary start gives (101) and the a.e. identity (102). Selecting its right-hand side as the density version, its margins equal one, and direct expansion using (2)–(3) gives

$$\int_I c^{\mathrm{mix}}_{s,r}(u,w)c^{\mathrm{mix}}_{r,t}(w,v)\,\mathrm{d}w = a_{s,t}c_{s,t}(u,v) + (1-a_{s,t}) = c^{\mathrm{mix}}_{s,t}(u,v).$$

for every u and a.e. v; integration in v gives exact kernel consistency for every target Borel set. The right derivative of (99) at $t = s+$ gives (103). $\square$

Theorem 4.2 (Exact strengthening of conditional mean reversion). *Suppose the base process has invariant mean $m = \int_E z\pi(\,\mathrm{d}z)$ and satisfies*

$$\mathbb{E}[Z_t - m \mid Z_s = x] = \mathcal{R}_{s,t}(x - m). \tag{104}$$

Then the reset-mixture process satisfies

$$\mathbb{E}[Y_t - m \mid Y_s = x] = a_{s,t}\mathcal{R}_{s,t}(x - m). \tag{105}$$

Its local drift is

$$b_{\mathrm{mix}}(t, x) = b_0(t, x) + \lambda(t)(m - x). \tag{106}$$

Thus a nonnegative reset intensity preserves the local sign condition and, for $\lambda > 0$, strictly accelerates the decay of conditional mean deviations away from m.

Proof. The base component of (99) contributes $a_{s,t}\mathcal{R}_{s,t}(x - m)$. A draw from π has centered mean zero, so the reset component contributes zero. This proves (105); its right derivative gives (106). □

Remark 4.3 (A precise theoretical advantage). The mixed model does not alter any one-time marginal under stationary start, but multiplies the base conditional-mean persistence by $a_{s,t}$. Hence $a_{s,t}\mathcal{R}_{s,t}$, the implied half-life, or the additive rate $-\partial_{t+} \log a_{s,t} = \lambda(t)$ are clear structural measures of its extra reversion channel. Whether this produces superior out-of-sample log scores or calibration is a separate empirical question.

4.2 Poisson realization

Theorem 4.4 (Independent-copy reset construction). *Let $\{Z^{(j)} : j \geq 0\}$ be independent copies of a stationary Markov process with transition evolution P and invariant law π. Let N be an independent nonhomogeneous Poisson process with intensity $\lambda(t)$, and set*

$$Y_t = Z_t^{(N_t)}. \tag{107}$$

Then Y is Markov, stationary in its one-time marginals, and has transition kernel (99) *and transition density* (100). *Under the stationary continuous-margin conditions of Theorem 4.1, it also has Copula density* (102). *A constant λ gives $a_{s,t} = e^{-\lambda(t-s)}$; a positive deterministic seasonal intensity is also admissible.*

Proof. Conditional on no jump in $(s, t]$, whose probability is $a_{s,t}$, the same copy evolves with kernel $P_{s,t}$. Conditional on at least one jump, the active copy at t is independent of the history through time s and has law π. The resulting conditional law is (99), whose density is (100), and it is a Markov evolution by Theorem 4.1. □

4.3 Risk-neutral bond pricing with reset risk

This section is a pricing theorem under a risk-neutral measure $\mathbb{Q}$, not a claim that a physical-measure stablecoin spot series identifies bond prices.

Theorem 4.5 (Nonlocal Feynman–Kac equation). *Under $\mathbb{Q}$, let the short rate be a stationary-reset OU process with generator*

$$\mathcal{G}h(y) = \kappa(m - y)h'(y) + \tfrac{1}{2}\sigma^2 h''(y) + \lambda \left\{ \int_{\mathbb{R}} h(z)\pi(\,\mathrm{d}z) - h(y) \right\}, \tag{108}$$

where $\pi = \mathcal{N}(m, \sigma^2/(2\kappa))$. Suppose the exponential discount functional is integrable and that the classical Feynman–Kac hypotheses hold; alternatively, assume conditions giving a unique *viscosity solution with the stated growth class. For jump generators and the corresponding nonlocal Feynman–Kac equations, see Cont and Tankov (2004, Chapters 8 and 12). Then*

$$g(t,y) = \mathbb{E}^{\mathbb{Q}}\left[\exp\left\{-\int_t^T Y_s \, \mathrm{d}s\right\} \middle| Y_t = y\right] \tag{109}$$

solves

$$\begin{cases} g_t + \kappa(m-y)g_y + \frac{1}{2}\sigma^2 g_{yy} + \lambda\{\int_{\mathbb{R}} g(t,z)\pi(\,\mathrm{d}z) - g(t,y)\} - yg = 0, \\ g(T,y) = 1. \end{cases} \tag{110}$$

Proof. Theorem 4.1 gives the generator (108). Apply the Feynman–Kac formula with killing rate y and terminal payoff one. Rearranging the generator equation yields (110). □

5 Empirical Analysis

5.1 Theory-to-empirics interface

We define a Markov process as mean reverting when it has an asymptotic mean and its infinitesimal conditional drift has the sign of the displacement toward that mean. It then constructs new processes through (i) monotone transformations of OU and CIR diffusions and (ii) mixtures of a consistent base copula family with the independence copula. Table 1 records the exact relationship between those constructions and the empirical implementation.

Sections 2–4 define processes and observable transition densities rather than an estimator. We therefore add maximum likelihood, composite likelihood, block bootstrap, PIT diagnostics, and rolling out-of-sample scores without altering the theoretical process definitions.

5.2 Data, state variables, and exact transition design

The raw file is Kraken's five-minute USDT/USD OHLCVT series. Its SHA-256 digest is

`417ea3aa71f86c4476c30568f251f3f873ae8b0d44c5f480f5ad2f35fc6acc22`.

The audit finds 797,178 observations from 29 March 2017 to 31 December 2025, 750,220 exact 300-second adjacent pairs, no duplicate timestamps, no missing fields, and grid coverage of 86.5186%. The main sample is 2021–2025. The training, validation, and test periods are 2021–2023, 2024, and 2025, respectively.

For OU, threshold-OU, MOU, transformed-OU, and semiparametric copula models the state is the log price

$$X_t = \log P_t, \qquad X_t = 0 \Longleftrightarrow P_t = 1. \tag{111}$$

Price CIR and MCIR instead use $P_t > 0$ directly. Depeg-pressure CIR uses $|P_t - 1| + 10^{-8}$ and is interpreted as an amplitude model, not a signed mean-reversion model. Transition pairs are formed by timestamp hashing: a pair (X_t, X_{t+h}) is used only when the timestamp difference is exactly h. Missing bars are never interpolated.

5.3 Empirical specifications and estimators

5.3.1 OU benchmark and finite-horizon drift

The OU benchmark is

$$\mathrm{d}X_t = \kappa(\theta - X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad \kappa > 0, \quad \sigma > 0. \tag{112}$$

Table 1: Correspondence between theory and empirical specifications

Theoretical object	Empirical counterpart	Interpretation
Mean-reversion definition	Five-minute local-linear estimate of $\mu_h(x) = h^{-1}\mathbb{E}[X_{t+h} - X_t \mid X_t = x]$ and sign statistic $\mathbf{1}\{(\theta - x)\mu_h(x) > 0\}$	Finite-horizon proxy; it does not prove the $h \downarrow 0$ limit or asymptotic moment convergence.
Stationary OU and CIR	Exact OU transition density for $X_t = \log P_t$; CIR transition density for P_t and for absolute depeg pressure	Model equations match. Price CIR is numerically weakly identified.
Linear transformed OU	$X_t = A_0 e^{(\kappa-d)t} r_t + B_0 e^{-dt}$ with $A_0 = 1$ and $B_0 = 0$	Exact restricted special case of the article's linear OU solution.
Linear transformed CIR	$P_t = A_0 e^{(\kappa-d)t} r_t + \theta[1 - A_0 e^{(\kappa-d)t}] + B_0 e^{-dt}$, with $A_0 = 1$ and $B_0 = 0$	Exact restricted formula; base CIR parameters are held fixed and the resulting model is weakly identified.
Mixed copula $C^{\mathrm{mix}}_{s,t} = a_{s,t} C_{s,t} + (1 - a_{s,t})\Pi$	MOU and MCIR with $a_{s,t} = e^{-\lambda(t-s)}$ and the stationary OU/Gamma marginal in the reset component	Exact constant-intensity case of the article's Poisson construction.
General $a_{s,t} = \exp\{-\int_s^t c(u)\,\mathrm{d}u\}$	Constant and deterministic seasonal $c(t)$ specifications	The finite-dimensional seasonal model is estimated in the full registry but does not determine model selection.
Gaussian copula with a general marginal	Gaussian and mixed Gaussian copulas with an empirical marginal	Copula construction is consistent, but the article's sufficient sign condition for mean reversion is not imposed and is not supported by the fitted finite-horizon drift.
Two-scale latent Gaussian copula	A weighted sum of fast and slow OU covariance components, optionally combined with an independence reset	Diagnostic extension: the two-dimensional latent state is Markov, but the observed scalar state is generally not one-dimensional Markov and is not an article proposition.
Nonlinear OU/CIR transformations	Exponential OU; quadratic and exponential CIR; and the stated special-function branches	All explicit families are implemented; weak identification, non-injectivity, monotonicity, and support failures are retained as empirical outcomes.
Mixed-copula bond-pricing PDE	Not used for stablecoin prices	Not applicable to the stated empirical objective.

Throughout the theory and empirical implementation, a positive restoring coefficient is written as $b(x) = \alpha(m - x)$; the opposite sign would describe mean repulsion.

Table 2: Sample construction

Sample	Period	Observations	Coverage (%)	Exact 5-minute pairs
Baseline	2021–2025	524,835	99.7998	524,610
Extended	2020–2025	627,782	99.4434	625,690
Full	2017–2025	797,178	86.5186	750,220
Training	2021–2023	314,759	99.8094	314,590
Validation	2024	105,153	99.7581	105,118
Test	2025	104,923	99.8126	104,900

Replication source: `output/tables/sample_summary.csv`.

For an interval h, its conditional mean and variance are

$$m_h(x) = \theta + e^{-\kappa h}(x - \theta), \tag{113}$$

$$v_h = \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa h}\right). \tag{114}$$

OU0 fixes $\theta = 0$, whereas OUF estimates θ. Parameters are estimated by exact Gaussian transition maximum likelihood. Ordinary Hessian standard errors, UTC-day cluster-sandwich standard errors, and UTC-day block bootstrap intervals are reported.

The model-free diagnostic estimates

$$\mu_h(x) = \frac{1}{h}\mathbb{E}[X_{t+h} - X_t \mid X_t = x] \tag{115}$$

by local-linear regression on a 99-point quantile grid. Bandwidth is selected by blocked cross-validation and uncertainty is computed from 1,000 UTC-day bootstrap replications. The sign-consistency ratio is

$$\mathrm{SCR}_h(\theta) = \frac{1}{G}\sum_{g=1}^{G} \mathbf{1}\{(\theta - x_g)\widehat{\mu}_h(x_g) > 0\}. \tag{116}$$

5.3.2 Parametric mixed-copula transition kernel

Let $P_h(x, \mathrm{d}y)$ be the transition kernel of a stationary base process with invariant distribution F. The article's constant-intensity Poisson construction gives

$$Q_h(x, \mathrm{d}y) = e^{-\lambda h}P_h(x, \mathrm{d}y) + \left(1 - e^{-\lambda h}\right)F(\mathrm{d}y). \tag{117}$$

For an OU base process, the conditional density used in estimation is therefore

$$q_h(y \mid x) = e^{-\lambda h}\phi(y; m_h(x), v_h) + \left(1 - e^{-\lambda h}\right)\phi\left(y; \theta, \frac{\sigma^2}{2\kappa}\right). \tag{118}$$

This is exactly the transition density implied by the article's process $Y_t = Y_t^{(N_t)}$ when the component processes are stationary OU processes. It nests the OU at $\lambda = 0$ and satisfies

$$\mathbb{E}[Y_{t+h} - \theta \mid Y_t = x] = e^{-(\kappa+\lambda)h}(x - \theta). \tag{119}$$

Hence the base diffusion half-life is $\log(2)/\kappa$, while the conditional mean half-life of the mixed process is $\log(2)/(\kappa + \lambda)$. These two quantities must not be conflated.

MOU0 fixes $\theta = 0$; MOUF jointly estimates $(\theta, \kappa, \sigma, \lambda)$. MCIR replaces the OU transition and Normal invariant density in Equation (118) with the CIR transition and its Gamma invariant density. Because the price-CIR likelihood is weakly identified, MCIR holds the base CIR parameters fixed and estimates only λ; MCIR is therefore a restricted diagnostic, not a full joint MLE.

5.3.3 Semiparametric multi-horizon copula

Let F_n be the empirical marginal CDF and $U_t = F_n(X_t)$. For a Gaussian base copula with $\rho_h = e^{-\kappa h}$, the fitted copula density is

$$c_h^{\text{mix}}(u, v) = e^{-\lambda h} c_{\text{G}}(u, v; \rho_h) + 1 - e^{-\lambda h}. \tag{120}$$

The parameters are estimated by an equally weighted composite likelihood over 5, 10, 30, 60, 120, and 360 minutes. Equation (120) defines a consistent stationary Markov copula construction. It does *not*, for an arbitrary F_n, automatically satisfy the sign restriction in the article's proposition for Gaussian copulas with general stationary marginals. This restriction is evaluated after estimation rather than assumed.

To test whether the failure at long horizons is caused by the single exponential correlation curve, define the two-scale correlation

$$\rho_h^{(2)} = w \exp(-\kappa_f h) + (1 - w) \exp(-\kappa_s h), \qquad 0 < w < 1, \quad 0 < \kappa_s < \kappa_f, \tag{121}$$

and the two-scale mixed density

$$c_h^{(2,\text{mix})}(u, v) = e^{-\lambda h} c_{\text{G}}(u, v; \rho_h^{(2)}) + 1 - e^{-\lambda h}. \tag{122}$$

Equation (121) is generated by the sum of two independent stationary OU factors. The latent vector is Markov, but its scalar sum is in general not one-dimensional Markov. Consequently, Equations (121)–(122) are used to diagnose multi-horizon dependence and density forecasts; they are not presented as a new instance of the article's one-dimensional Markov theorem. Four models are fitted on exactly the same deterministic subsample at each horizon: SingleGaussian, SingleMixed, TwoScaleGaussian, and TwoScaleMixed.

5.3.4 Prespecified model hierarchy

The hierarchy is fixed before interpreting the additional results. OUF and MOUF are the *core structural comparison*, because their transition laws directly match the article. Heteroskedastic threshold OU is the *main empirical extension* for conditional asymmetry. The four empirical-marginal copula models are *diagnostic extensions*. Random walk and nested-OU forecasts are *prediction benchmarks*. CIR, MCIR, and affine transformed models are *appendix feasibility checks*. A diagnostic model can reveal a missing dependence scale without inheriting the theoretical status of the core models.

Replication source: `output/tables/article_model_hierarchy.csv`.

5.3.5 Model comparison and inference

Likelihood, AIC, and BIC are compared only for models fitted to the same state, sample, and likelihood. PIT uniformity and Ljung–Box tests assess transition density calibration. Because $\lambda = 0$ is a boundary point, the ordinary χ^2 likelihood-ratio reference distribution is not used. The primary parametric bootstrap uses the same deterministic 25,000 pairs for the observed and simulated statistics, holds the observed regressors fixed under the null, and jointly re-estimates $(\theta, \kappa, \sigma, \lambda)$ under MOUF in every replication. Out-of-sample comparison uses negative log score (NLS), CRPS, MAE, RMSE, Brier score, event log score, AUC, and calibration. All 2025 predictions use parameters selected without 2025 outcomes.

5.4 Empirical results

5.4.1 Nonparametric evidence and OU benchmark

The point estimates predominantly have the correct direction, but the simultaneous confidence band in Figure 1 contains zero over a large part of the state grid. The evidence is therefore directional rather than a proof of pointwise significant infinitesimal mean reversion.

Table 3: Finite-horizon mean-reversion sign diagnostics

Anchor definition	$\widehat{\theta}$	SCR	95% bootstrap interval	Grid points
Peg anchor	0	0.7778	[0.7677, 0.8788]	99
Sample anchor	6.700×10^{-5}	0.8788	[0.8586, 0.9697]	99

Replication source: `output/tables/sign_test_baseline.csv`.

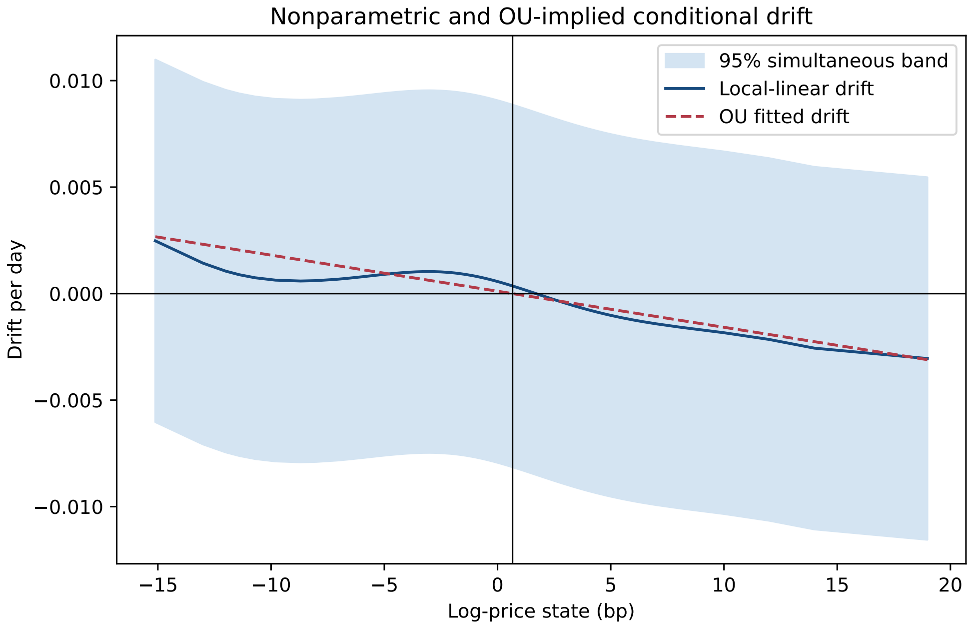


Figure 1: Local-linear finite-horizon drift and the OUF-implied drift.

Table 4: OU, asymmetric OU, and MOU parameter estimates

Model	$10^4\widehat{\theta}$	$\widehat{\kappa}$	$\widehat{\lambda}$	$\widehat{\sigma}$	Base HL (min)	Effective HL (min)
OUF	0.665	1.6932	—	0.001355	589.5	589.5
MOUF	7.248	0.02484	1.8863	0.000963	40,190.4	522.3

Rates are per day. OUF UTC-day cluster standard errors are $(0.166, 0.2099, 0.0001456)$ for $(10^4\theta, \kappa, \sigma)$. MOUF cluster standard errors are $(6.496, 0.01419, 0.18835, 0.00000950)$ for $(10^4\theta, \kappa, \lambda, \sigma)$. The large difference between the OUF and MOUF estimates of θ is a warning against interpreting the MOUF stationary mean as the economic peg without further stationarity restrictions.

Replication source: `output/tables/model_estimates.csv`.

The heteroskedastic threshold OU estimates $(\kappa_+, \kappa_-) = (1.5674, 1.8103)$ and $(\sigma_+, \sigma_-) = (0.001248, 0.001515)$. Its likelihood gain is driven mainly by asymmetric conditional volatility rather than by a large difference in reversion speed.

5.4.2 Does the mixed kernel add explanatory power?

Table 5: Comparable five-minute transition likelihoods

Model	Parameters	Log likelihood	AIC	BIC
OU0	2	4,207,015.02	$-8,414,026.04$	$-8,414,003.70$
OUF	3	4,207,021.25	$-8,414,036.50$	$-8,414,002.98$
Threshold OU, heteroskedastic	4	4,211,719.53	$-8,423,431.07$	$-8,423,386.38$
MOU0	3	4,351,515.34	$-8,703,024.68$	$-8,702,991.17$
MOUF	4	4,351,562.04	$-8,703,116.07$	$-8,703,071.39$

Replication source: `output/tables/model_comparison.csv`.

Relative to OUF, MOUF increases the total log likelihood by 144,540.79, or approximately 0.2755 per transition pair, and lowers AIC by 289,079.58. This is the clearest in-sample evidence for the mixed kernel. The gain reflects a two-scale transition density: a highly persistent narrow base component and a small-probability draw from a much wider invariant component.

The primary boundary bootstrap removes the design mismatch in the earlier diagnostic. Both the observed and null-simulated statistics use the same deterministic systematic sample of 25,000 out of 524,610 exact transitions, the same fixed regressor values and five-minute step, the same OUF refit, and the same joint MOUF optimization. The observed statistic is $LR =$ 12,722.80. Among $B = 500$ null replications, none is as large (maximum 0.635), giving the finite-replication correction

$$\widehat{p} = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\{LR_b \geq LR_{\text{obs}}\}}{B+1} = \frac{1}{501} = 0.001996. \tag{123}$$

The alternative optimizer reports convergence in 98.2% of replications and $\widehat{\lambda}$ is at its zero boundary in 99.4%, as expected for this nonregular null. Thus the ordinary χ^2 approximation is inappropriate, but the matched conditional bootstrap provides strong evidence that the MOUF density adds a component absent from OUF on this prespecified design. The claim remains conditional on the 25,000-pair systematic design and the maintained OU null; it is not a proof that the economic data-generating process is MOUF.

Replication source: `output/tables/matched_joint_mou_lr_bootstrap.csv`.

Replication source: `output/tables/matched_joint_mou_lr_bootstrap_draws.csv`.

The former $B = 200$ profiled calculation is retained only as a historical sensitivity file because it compares a full-sample observed statistic with shorter simulated paths and does not jointly fit the same alternative. It is not used for the primary significance statement.

Replication source: `output/tables/mixed_lr_bootstrap.csv`.

Three reported reset intensities answer different questions: These estimates are not interchangeable. Their dispersion is direct evidence that κ and λ are sensitive to the marginal model, estimation objective, and horizon set.

5.4.3 Multi-horizon dependence

The mixed term reduces absolute dependence error at 5 and 10 minutes, but makes the fit worse from 30 minutes onward. At 360 minutes the data remain much more persistent than either single-exponential specification. A constant-intensity independence reset therefore does not replace a slow latent factor, a regime switching model, or a multi-scale OU structure.

Table 6: Reset-intensity estimates under different objectives

Estimation objective	$\widehat{\lambda}$ per day	Interpretation
Joint five-minute MOUF likelihood	1.8863	Jointly changes the OU base and reset component
Matched joint MOUF likelihood	1.9185	Same 25,000-pair design as the boundary bootstrap
Restricted boundary profile	0.7678	Profiles λ conditional on a re-estimated OU base
Multi-horizon empirical-marginal copula	0.2557	Composite dependence fit over 5–360 minutes

Table 7: Empirical and fitted Gaussian-rank dependence

h (min)	Pairs	Empirical	Base fit	Mixed fit	Reset weight
5	524,610	0.99128	0.99692	0.99603	0.00089
10	524,494	0.98868	0.99384	0.99208	0.00177
30	524,279	0.98067	0.98165	0.97643	0.00531
60	524,143	0.97160	0.96363	0.95342	0.01060
120	524,017	0.95664	0.92858	0.90901	0.02108
360	523,734	0.91630	0.80068	0.75111	0.06192

Replication source: `output/tables/copula_horizon_fit.csv`.

The prespecified two-scale comparison confirms that the missing persistence is a separate dimension from the reset mixture.

Table 8: Single- versus two-scale empirical-marginal copula fits

Model	$\widehat{\kappa}_s$	$\widehat{\kappa}_f$	$\widehat{w}$	$\widehat{\lambda}$	Mean log score	Dependence RMSE
SingleGaussian	1.1398	—	—	0	1.42051	0.07022
SingleMixed	0.8889	—	—	0.2505	1.46991	0.07018
TwoScaleGaussian	0.3348	251.0089	0.01244	0	1.52663	0.00377
TwoScaleMixed	0.1361	225.0237	0.01215	0.5078	1.59503	0.03223

Rates are per day. "Mean log score" is the equally weighted mean copula log density over six horizons, using at most 100,000 deterministic pairs per horizon; higher is better. Dependence RMSE compares the six empirical Gaussian-rank correlations with the fitted correlations; lower is better. All four optimizers converged, and repeated starts attained the same best solution.

Replication source: `output/tables/copula_scale_model_comparison.csv`.

The two metrics answer different questions. TwoScaleGaussian reduces dependence RMSE by 94.6% relative to SingleGaussian and reproduces the 360-minute dependence as 0.9083 versus the empirical 0.9163. TwoScaleMixed has the highest full-density score, improving it by 0.0684 over TwoScaleGaussian, but its reset component lowers the fitted 360-minute dependence to 0.8410 and therefore increases correlation RMSE. Thus the slow factor explains the long-memory shape, whereas the mixed component explains additional features of the joint density; a likelihood gain alone is not evidence of a better correlation curve.
Replication source: `output/tables/copula_scale_horizon_fit.csv`.

For the empirical marginal F_n, the fitted semiparametric conditional mean has five-minute sign consistency of only 34.34% around the exact peg and 42.42% around the empirical marginal mean 6.713×10^{-5}. Thus the empirical- marginal Gaussian/mixed-copula fit is useful as a

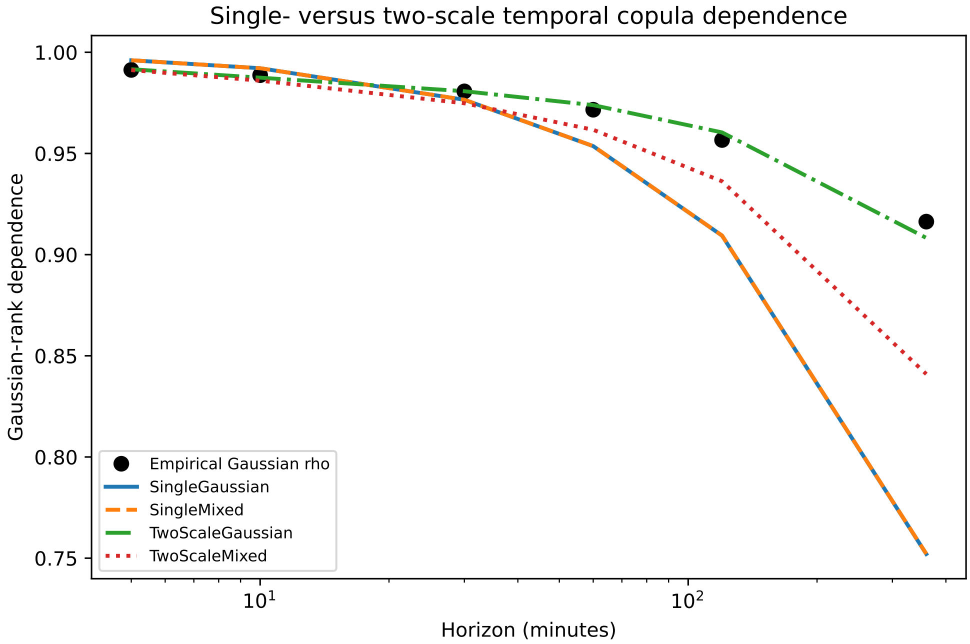
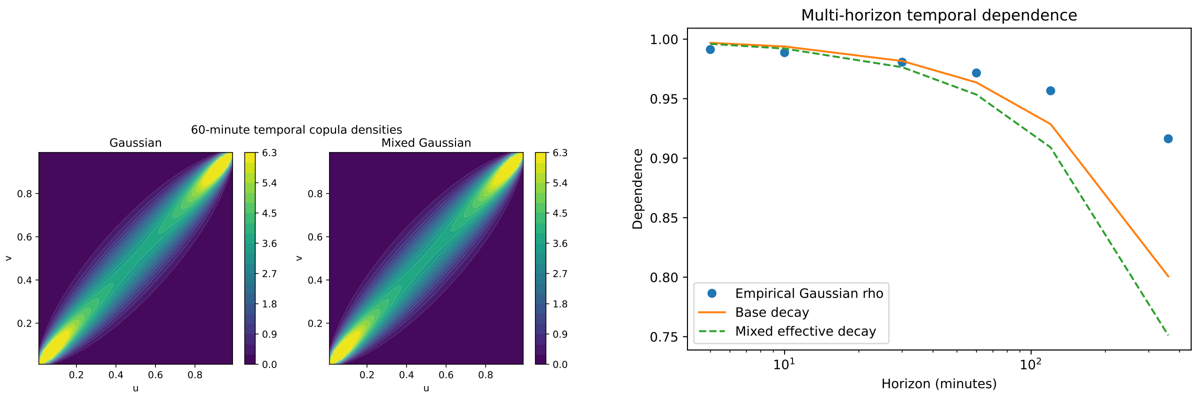


Figure 2: Empirical multi-horizon dependence and the four scale/reset specifications.



(a) Gaussian and mixed copula contours.

(b) Dependence across horizons.

Figure 3: Copula shape and multi-horizon dependence.

dependence diagnostic, but it is not empirically verified as a mean-reverting process under the article's definition. This conclusion is distinct from the parametric MOU result, whose Normal invariant marginal and OU base satisfy the theoretical construction by design.

Replication source: `output/tables/copula_conditional_drift_baseline.csv`.

Replication source: `output/tables/descriptive_statistics_baseline.csv`.

5.4.4 Density diagnostics and out-of-sample performance

Numerical optimization succeeds for OUF and MOUF, but both models are rejected by PIT uniformity and by Ljung–Box tests of transformed residuals and squared residuals (reported numerical p-values are effectively zero). The mixed model therefore improves relative likelihood without achieving absolute transition- density adequacy.

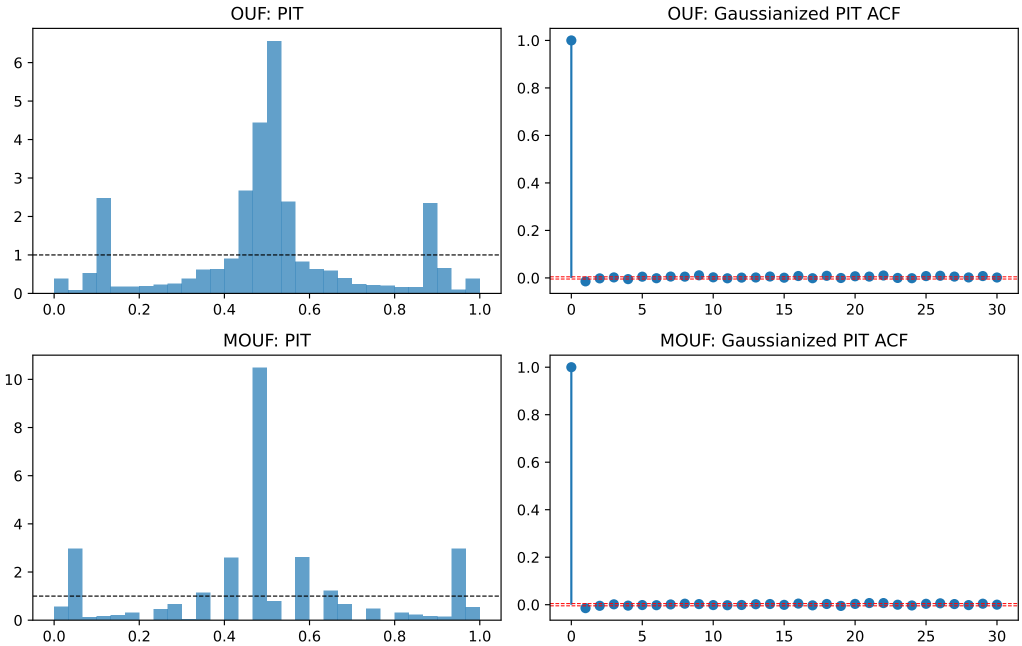


Figure 4: PIT and residual diagnostics for OUF and MOUF.

Table 9: Validation-period model selection, 2024

Model	Mean NLS	Mean CRPS	Mean RMSE	NLS rank
OUF	−7.18759	0.00013091	0.00023667	1
Random walk	−7.13145	0.00013279	0.00021416	2
Nested-OU proxy	−7.13006	0.00014465	0.00026387	3
MOUF	−7.10424	0.00015907	0.00029675	4

Replication source: `output/tables/validation_model_selection.csv`.

Lower NLS and CRPS are better. OUF is selected on the validation period. In the 2025 test period, MOUF has a better 5-minute NLS for the 50-basis-point depeg specification (-8.7632 versus -8.7025 for OUF), but OUF is better at 60, 360, and 1,440 minutes. The mixed kernel therefore has a localized short-horizon density advantage rather than uniform out-of-sample dominance.

Replication source: `output/tables/forecast_scores.csv`.

The scale/reset comparison is a separate copula-density exercise. The empirical marginal and all copula parameters are estimated using 2021–2023 only; the fitted objects are then frozen for 2024 and 2025. Table 10 reports the equally weighted mean across the six horizons.

Table 10: Out-of-sample copula log scores by model scale

Model	Training 2021–2023	Validation 2024	Test 2025
SingleGaussian	1.27385	1.54637	1.38939
SingleMixed	1.31164	1.59861	1.43596
TwoScaleGaussian	1.40784	1.61073	1.50638
TwoScaleMixed	1.47899	1.66640	1.57116

Higher log score is better. Each period-specific mean first averages over observations within a horizon and then gives equal weight to each of the six horizons. No validation or test outcomes enter parameter estimation.

Replication source: `output/tables/copula_scale_oos_scores.csv`.

TwoScaleMixed exceeds SingleMixed by 0.06780 log-score units in validation and 0.13520 in test; it also exceeds TwoScaleGaussian by 0.05567 and 0.06478. This is the clearest out-of-sample metric in favor of combining a slow factor with the mixed reset. The advantage is not uniform by horizon: on the 2025 test data the two-scale models score worse than SingleMixed at 5 and 10 minutes, but better at 30, 60, 120, and 360 minutes. The result supports multi-scale dependence, not an unqualified claim that the mixed process dominates every forecast horizon.

The baseline 50/25-basis-point event rule identifies eight events. Their empirical recovery-time median is 155 minutes, compared with 665 minutes under OUF simulation and 1,085 minutes under MOUF simulation. The reset mechanism does not necessarily move the state toward the peg; it draws from the invariant marginal. Consequently, the effective conditional-mean half-life in Equation (119) is not a first-passage-time statistic.
Replication source: `output/tables/first_passage_comparison.csv`.

5.4.5 Frequency, sample, and block-length robustness

Table 11: Baseline frequency robustness

Minutes	Exact pairs	$\widehat{\kappa}$	$\widehat{\lambda}$	OU HL	Effective HL
5	524,610	1.6932	0.6969	589.5	417.6
15	174,578	1.8504	0.1784	539.4	492.0
30	87,068	1.3992	0.1323	713.4	651.7
60	43,322	1.1311	0.0785	882.5	825.2

All non-five-minute rows use complete aggregation bins only. Half-lives are in minutes. In this table λ is estimated conditionally on fast OU parameters, so it should not be compared numerically with the joint MOUF estimate without qualification.

Replication source: `output/tables/robustness_estimates.csv`.

The decline in $\widehat{\kappa}$ at lower sampling frequencies is consistent with microstructure noise inflating high-frequency reversion estimates. The 1-, 3-, and 7-day moving-block bootstrap standard errors for OUF κ are 0.2021, 0.2300, and 0.2622, respectively; uncertainty increases with block length but the reversion-rate estimate remains positive. In the full 2017–2025 sample, retaining three flagged hard anomalies produces $\widehat{\kappa} = 303.09$, whereas excluding them gives $\widehat{\kappa} = 2.742$. Structural interpretation of the full-sample high-frequency estimate is therefore untenable without explicit outlier treatment.
Replication source: `output/tables/ou_block_length_robustness.csv`.

5.5 Full-model registry and validation selection

The second-stage exercise removes the MVP complexity screen. Every article model with an explicit finite-dimensional observation density, and every benchmark or marginal–copula family required by the empirical methodology, is either estimated or assigned a terminal failure diagnostic before any model is chosen for emphasis. The registry contains 25 theory/methodology objects. Of these, 18 are estimated in the current or pre-existing pipeline, five terminate with a retained monotonicity, support, or endpoint failure, the stationary Gaussian–Gamma construction is estimable but fails the article's shape condition, and the bond-pricing PDE is retained as not observable from spot data.

Replication source: `output/tables/full_model_implementation_status.csv`.

For every comparable model, selection uses the same 2024 five-minute conditional log score on the log-price measure,

$$\widehat{S}_m^V = \frac{1}{N_V}\sum_{i=1}^{N_V} \log q_m(x_{t_i} \mid \mathcal{F}_{t_i-}), \qquad N_V = 40{,}000. \tag{124}$$

Price-density models are converted by the Jacobian $q_X(x_t \mid x_{t-h}) = q_P(e^{x_t} \mid e^{x_{t-h}})e^{x_t}$. Copula models use $q_X(x_t \mid x_{t-h}) = c_h(F(x_{t-h}), F(x_t))f(x_t)$. Thus AIC values from ordinary and composite likelihoods do not enter this cross-class ranking. For the two-state OU, the primary one-step score is the exact hidden-state predictive density

$$q_t(x_t \mid \mathcal{F}_{t-1}) = \sum_{j=1}^{2}\left(\sum_{i=1}^{2}\alpha_{t-1,i}p_{ij}\right)p_j(x_t \mid x_{t-1}), \tag{125}$$

with the filter reset to its stationary probabilities after every timestamp gap. Longer-horizon Markov-switching entries are labelled as endpoint-mixture approximations and do not determine selection.

Table 12: Unified 2024 validation ranking at the five-minute horizon

Rank	Model	Mean log score	Coverage	Identification audit
1	Markov-switching OU	8.824686	1.000	weak: slow regime/boundary
2	Student-t margin + Student-t copula	8.781202	1.000	weak: $\nu = 2.05$ grid boundary
3	Student-t margin + mixed Student-t copula	8.781201	1.000	weak: $\lambda \simeq 0$, ν boundary
23	MOUF	8.431420	1.000	passes numerical identification gate
43	OUF	8.221318	1.000	passes numerical identification gate
44	Random walk	8.220764	1.000	no-reversion benchmark

The rank numbers for MOUF, OUF, and the random walk are their positions in the complete candidate table; omitted rows are not omitted from estimation or ranking. Selection uses 2024 only. The 2025 scores are post-extension exploratory because aggregate 2025 outcomes had been inspected previously.

Replication source: `output/tables/full_model_validation_ranking.csv`.

The selected model's advantage is not merely a point-ranking artifact. Day-block resampling of paired score differences gives the comparisons in Table 13. These intervals measure predictive-score uncertainty conditional on the frozen estimates; they do not repair weak identification of the switching parameters.

The expanded grid also makes the mixed-copula claim more precise. With the Student-t marginal fixed, adding the independent-reset component raises the Gaussian validation score by 0.078347 and the two-scale Gaussian score by 0.048432. It changes the Student-t copula score by only -6.95×10^{-7} because $\widehat{\lambda}$ is effectively zero. For Frank, Clayton, survival Clayton, Gumbel,

Table 13: Paired day-block comparison of the selected model

Reference	$\widehat{S}_{MSOU} - \widehat{S}_{ref}$	95% block interval	Daily win share
Student-t margin + Student-t copula	0.043484	[0.018242, 0.065954]	0.656
MOUF	0.393266	[0.355106, 0.429977]	0.915
OUF	0.603368	[0.566379, 0.640416]	0.956
Random walk	0.603922	[0.564066, 0.642485]	0.959

Replication source: `output/tables/full_model_pairwise_validation.csv`.

and survival Gumbel, the fitted mixed versions hit weak-identification/boundary diagnostics and reduce the validation score by 1.98 to 3.21. The empirical advantage is therefore specific to the Gaussian and two-scale Gaussian reset constructions; it is not a theorem that mixing improves every copula family.

Replication source: `output/tables/full_mixture_incremental_scores.csv`.

The transformation audit is equally informative. Affine OU is estimable; exponential OU converges but is weakly identified. The OU special-function solution is even in the latent state and hence non-injective on $\mathbb{R}$. Affine CIR is estimated but weak, while quadratic CIR, exponential CIR, and the prespecified CIR special-function branches fail global monotonicity or observed support. These failures are substantive outcomes, not silent exclusions. The Gamma Case 1 equation has no finite shape root on the audited range, so Case 2 lacks the stationary endpoint it requires. Finally, the mixed-copula bond PDE cannot be estimated without a bond-price cross section and a risk-neutral measure; its equation is implemented only as a non-observable theoretical object in this spot-price application.

5.6 Consistency between the theory and the empirical evidence

The first point of agreement is directional. USDT/USD log prices display finite-horizon reversion around a near-zero log-price anchor, consistent with the sign restriction in Section 2. The OU process therefore supplies an interpretable local benchmark. Its rejected transition diagnostics, however, show that a correct restoring direction does not imply a correct conditional density. This is precisely the distinction made by the theory between mean reversion and distributional specification.

The mixed-copula results provide a second, more specific correspondence. For stationary OU and CIR bases, the empirical density is exactly the constant-intensity Poisson-reset kernel of Section 4. For the OU base, the matched-design boundary experiment rejects the no-reset restriction at the minimum attainable corrected level 1/501. The result is evidence for a two-component transition density—a narrow persistent component together with rare wide transitions—and not causal identification of λ with redemption, arbitrage, or news. Consistently with that qualification, MOUF does not dominate OUF in the original validation exercise, PIT diagnostics, or empirical first-passage times.

The multi-horizon evidence identifies a different omission. Adding a slow latent factor reduces the six-horizon dependence RMSE from 0.07022 to 0.00377. Adding an independent reset to the two-scale model raises validation and test copula log scores to 1.66640 and 1.57116, but increases the correlation RMSE to 0.03223. Hence persistence shape and density score are distinct targets. This finding is also theoretically coherent: the two-factor latent process is Markov in its augmented state, whereas its scalar observation is not generally a one-dimensional Markov process of the class developed in Sections 2–4. The empirical-marginal copula likewise remains a dependence and forecasting extension unless the marginal sign restriction is imposed during estimation.

The transformation results reinforce the role of the admissibility conditions in Section 3. Every explicit finite-dimensional family is either estimated or assigned a terminal diagnosis.

Nonlinear CIR variants fail monotonicity or support, exponential OU is weakly identified, and the OU special-function map is non-injective. These outcomes do not contradict the transformation equations; they show why a formal solution is insufficient for an observable one-dimensional density. Among these families, only the affine-OU calibration is suitable for routine empirical comparison.

Finally, the common validation rule separates structural contribution from forecast ranking. The two-state Markov-switching OU has the largest 2024 five-minute log score and a positive paired advantage over the Student-t copula benchmark, but its near-zero slow-regime rate is weakly identified. The most defensible mixed-copula claim is therefore narrower and sharper: relative to an OU transition, the Poisson-reset kernel preserves the invariant marginal and Markov semigroup while materially improving the matched in-sample density; within Gaussian and two-scale Gaussian dependence, it also improves validation and test copula scores. The reset is redundant for the fitted Student-t copula, performs poorly for the fitted Archimedean mixtures, and does not alone generate the observed slow persistence. The mixed copula is thus the paper's principal theoretical innovation, whereas the forecasting leader is determined by the frozen empirical comparison.

6 Conclusion

This paper develops a rigorous copula-based theory of mean-reverting Markov processes and examines its empirical relevance for USDT/USD. The theory makes two distinctions that are essential in applications. First, an invariant distribution or a decaying dependence coefficient is not, by itself, a directional mean-reversion result; the conditional drift must point toward the target. Second, a finite-dimensional formula is not an observable model unless its transformation is invertible on the relevant support and its transition density is normalized. These requirements preserve the valid affine and monotone OU/CIR constructions while explaining the empirical failure of several quadratic, exponential, and special-function branches.

The mixed-copula construction is the main theoretical result. A multiplicative mixture of a stationary transition copula and independence defines a consistent Markov semigroup, leaves the invariant marginal unchanged, and has a Poisson reset representation. Its conditional centered mean is the base conditional mean multiplied by the no-reset probability, so the local generator acquires the transparent restoring term $\lambda(m - x)$. This provides a parsimonious way to alter transition density shape and recovery speed without rebuilding the marginal distribution.

The stablecoin evidence is consistent with the motivation for that separation, but it also defines its limits. USDT/USD exhibits finite-horizon directional reversion, while a single OU transition fails density diagnostics. The mixed-OU kernel is strongly favored in the matched in-sample boundary experiment, and reset mixtures improve out-of-sample log scores for Gaussian and two-scale Gaussian dependence. The improvement is not generic across copula families, and a constant reset rate cannot reproduce the observed slow multi-horizon persistence. The strongest five-minute forecasting model is instead a two-state OU with weakly identified slow-regime dynamics. Accordingly, the evidence supports the mixed copula as a structurally coherent density extension, not as a universally dominant forecasting law or a uniquely identified model of redemption and arbitrage.

Several extensions follow naturally. Economic interpretation of reset risk requires covariates for liquidity, redemptions, reserve news, and market stress. Slow persistence calls for an explicitly augmented Markov state rather than a horizon-by-horizon scalar copula approximation. Finally, the nonlocal pricing equation should be taken to bond or derivative data under a specified risk-neutral measure; it is not identified by a spot stablecoin series alone. These extensions preserve the paper's central principle: marginal behavior, temporal dependence, mean-reversion direction, and predictive performance must be modeled and tested separately.

A Detailed proofs

This appendix expands the measure-theoretic steps that are easy to hide inside phrases such as "interchange the limit and the conditional expectation." It covers the general results used repeatedly in the main text. The model-specific OU, CIR, and special-function calculations remain next to their statements because they are direct substitutions into the displayed differential equations.

A.1 Copula kernels, density versions, and partial derivatives

Detailed proof of the kernel and derivative claims in Proposition 2.2. Fix $s < t$ and suppress the time subscripts temporarily. For the selected jointly Borel density c, define

$$K(u,B) = \int_B c(u,v)\,\mathrm{d}v, \qquad H^c(u,y) = K(u,(0,y]) = \int_0^y c(u,v)\,\mathrm{d}v.$$

Row normalization makes $B \mapsto K(u,B)$ a probability measure for every u. Joint measurability of c, first for intervals and then by the monotone class theorem for all $B \in \mathcal{B}(I)$, makes $u \mapsto K(u,B)$ Borel. Column normalization and Tonelli's theorem give, for every Borel B,

$$\int_I K(u,B)\,\mathrm{d}u = \int_B \left\{ \int_I c(u,v)\,\mathrm{d}u \right\} \mathrm{d}v = \mathrm{Leb}(B), \tag{126}$$

so the selected kernel preserves the uniform law.

For every $x, y \in [0,1]$, the copula reconstructed from c satisfies

$$C(x,y) = \int_0^x H^c(a,y)\,\mathrm{d}a. \tag{127}$$

For fixed y, the function $H^c(\cdot,y)$ is measurable and bounded by one. The fundamental theorem for Lebesgue integrals therefore implies

$$\partial_1 C(u,y) = H^c(u,y) \quad \text{for Leb-a.e. } u. \tag{128}$$

This is an equality for each fixed y; its exceptional set may initially depend on y. Because $H^c(u,\cdot)$ and every conditional-CDF version are right-continuous, equality at all rational y outside one null set extends to all $y \in [0,1]$ outside that same source-state null set.

At any source state u where $a \mapsto H^c(a,y)$ is continuous, the pointwise upgrade follows directly from the integral difference quotient:

$$\begin{aligned} \frac{C(u+h,y) - C(u,y)}{h} - H^c(u,y) &= \frac{1}{h} \int_u^{u+h} \{H^c(a,y) - H^c(u,y)\}\,\mathrm{d}a, \\ \left| \frac{C(u+h,y) - C(u,y)}{h} - H^c(u,y) \right| &\le \sup_{|a-u| \le |h|} |H^c(a,y) - H^c(u,y)| \longrightarrow 0. \end{aligned} \tag{129}$$

Thus a continuity condition on the integrated density row, not merely existence of a Leb^2-a.e. density, is what upgrades (128) to a pointwise derivative identity. The modified partial Dini construction of Fang et al. (2020) supplies another pointwise CDF-valued version $\mathfrak{D}_1 C(u,\cdot)$ satisfying the same reconstruction formula. Applying (128) to both versions shows that $\mathfrak{D}_1 C(u,\cdot) = H^c(u,\cdot)$ for a.e. u, although equality can fail on a source-null set. These uses of absolute continuity and differentiation are instances of the results in Folland (1999, Chapter 3).

Now fix $0 \le s < r < t$ and $u \in I$. Since (3) holds for a.e. v, equality of its integrals over an arbitrary $B \in \mathcal{B}(I)$ and Tonelli's theorem give

$$\begin{aligned} K_{s,t}(u,B) &= \int_B \int_I c_{s,r}(u,w) c_{r,t}(w,v)\,\mathrm{d}w\,\mathrm{d}v \\ &= \int_I c_{s,r}(u,w) \left\{ \int_B c_{r,t}(w,v)\,\mathrm{d}v \right\} \mathrm{d}w \\ &= \int_I K_{r,t}(w,B) K_{s,r}(u,\,\mathrm{d}w). \end{aligned} \tag{130}$$

This proves the kernel Chapman–Kolmogorov equation for every selected source u and every Borel B. Conversely, if both sides are absolutely continuous in the target variable, equality for every B identifies their Radon–Nikodym densities only for a.e. v.

Finally, let $C = C_{s,r}$ and $D = C_{r,t}$. The functions

$$w \longmapsto \int_0^x c_{s,r}(a,w)\,\mathrm{d}a, \qquad w \longmapsto \int_0^y c_{r,t}(w,b)\,\mathrm{d}b$$

are measurable a.e. representatives of $\partial_2 C(x,w)$ and $\partial_1 D(w,y)$, respectively. Substitution into (9), followed by Tonelli and density CK, gives

$$\begin{aligned} (C_{s,r} * C_{r,t})(x,y) &= \int_I \int_0^x c_{s,r}(a,w)\,\mathrm{d}a \int_0^y c_{r,t}(w,b)\,\mathrm{d}b\,\mathrm{d}w \\ &= \int_0^x \int_0^y \left\{ \int_I c_{s,r}(a,w) c_{r,t}(w,b)\,\mathrm{d}w \right\} \mathrm{d}b\,\mathrm{d}a \\ &= \int_0^x \int_0^y c_{s,t}(a,b)\,\mathrm{d}b\,\mathrm{d}a = C_{s,t}(x,y). \end{aligned} \tag{131}$$

All changes in integration order are justified by nonnegativity. Conversely, equality of these copula CDFs identifies their two-dimensional densities only Leb^2-a.e. and hence, by Fubini, yields conditional density composition only for a.e. source state. This proves why star-product consistency alone is weaker than the selected pointwise kernel condition in Assumption 2.1. □

A.2 Short-time kernels acting on moving functions

Lemma A.1 (Moving-function kernel lemma)**.** *Fix $t \ge 0$ and an interior state $u \in I$. Write*

$$K_h(u,\,\mathrm{d}v) = c_{t,t+h}(u,v)\,\mathrm{d}v, \qquad h > 0,$$

so that $K_h(u,I) = 1$. The following statements hold.

(a) Suppose that

$$K_h\big(u, \{v : |v-u| \ge \delta\}\big) \longrightarrow 0 \quad \textit{for every } \delta > 0. \tag{132}$$

If g is bounded and continuous at u, then

$$\int_I g(v) K_h(u,\,\mathrm{d}v) \longrightarrow g(u). \tag{133}$$

(b) In addition to part (a), let $r_h : I \to \mathbb{R}$ be measurable and suppose g is bounded and continuous at u and

$$\|r_h - g\|_\infty \longrightarrow 0. \tag{134}$$

Then

$$\int_I r_h(v) K_h(u,\,\mathrm{d}v) \longrightarrow g(u). \tag{135}$$

(c) *For an unbounded g, let $w : I \to [1, \infty)$ be measurable and define $\|f\|_w = \sup_{v \in I} |f(v)|/w(v)$. If*

$$\|r_h - g\|_w \longrightarrow 0, \tag{136}$$

$$\sup_{0<h<h_0} \int_I w(v) K_h(u, \mathrm{d}v) < \infty, \tag{137}$$

$$\int_I g(v) K_h(u, \mathrm{d}v) \longrightarrow g(u), \tag{138}$$

then (135) *still holds.*

Proof. For part (a), split the integral at distance δ from u:

$$\begin{aligned}\left| \int_I g(v) K_h(u, \mathrm{d}v) - g(u) \right| &\le \int_I |g(v) - g(u)| K_h(u, \mathrm{d}v) \\ &\le \sup_{|v-u|<\delta} |g(v) - g(u)| + 2\|g\|_\infty K_h\big(u, \{|v - u| \ge \delta\}\big).\end{aligned}$$

First let $h \downarrow 0$ and use (132); then let $\delta \downarrow 0$ and use continuity of g at u. This proves (133), or equivalently $K_h(u, \cdot) \Rightarrow \delta_u$.

For part (b), row normalization gives

$$\begin{aligned}\left| \int_I r_h(v) K_h(u, \mathrm{d}v) - g(u) \right| &\le \int_I |r_h(v) - g(v)| K_h(u, \mathrm{d}v) \\ &\quad + \left| \int_I g(v) K_h(u, \mathrm{d}v) - g(u) \right| \\ &\le \|r_h - g\|_\infty + \left| \int_I g(v) K_h(u, \mathrm{d}v) - g(u) \right|.\end{aligned}$$

Both terms tend to zero by (134) and part (a).

For part (c), replace the first term in the preceding bound by

$$\int_I |r_h(v) - g(v)| K_h(u, \mathrm{d}v) \le \|r_h - g\|_w \int_I w(v) K_h(u, \mathrm{d}v).$$

Equations (136)– (138) prove the claim. This weighted version is the relevant one for Normal, Student, or other quantiles that diverge at the endpoints of I. ☐

Pointwise convergence $r_h(v) \to g(v)$ alone is not enough in this lemma: the measure $K_h(u, \mathrm{d}v)$ also changes with h. Conditions (134) or (136)–(137) provide the uniform integrability needed for the moving integrand.

A.3 Full integral proof of the quantile–generator drift identity

Detailed proof of Proposition 2.22. Fix t and put $q_t(v) = q(t, v) = Q_t(v)$ and $\dot{q}_t(v) = \partial_t q(t, v)$. Assume that all displayed first moments are finite and that, for the state u under consideration,

$$\mathcal{A}_t q_t(u) = \lim_{h \downarrow 0} \frac{\int_I q_t(v) c_{t,t+h}(u, v)\, \mathrm{d}v - q_t(u)}{h} \tag{139}$$

exists. Also assume either part (b) or part (c) of Lemma A.1 with

$$r_h(v) = \frac{q(t + h, v) - q(t, v)}{h}, \qquad g(v) = \dot{q}_t(v). \tag{140}$$

These are explicit sufficient conditions for the limit-interchange clause in Proposition 2.22.

For a regular conditional kernel version, strict invertibility of q_t gives, for Leb-almost every u,

$$P^X_{t,t+h}\big(q_t(u), B\big) = \int_I \mathbf{1}_B\big(q_{t+h}(v)\big) c_{t,t+h}(u,v)\,\mathrm{d}v. \tag{141}$$

Consequently,

$$b_X(t, q_t(u)) = \lim_{h\downarrow 0} \frac{\int_I q_{t+h}(v) c_{t,t+h}(u,v)\,\mathrm{d}v - q_t(u)}{h}. \tag{142}$$

Using $\int_I c_{t,t+h}(u,v)\,\mathrm{d}v = 1$, the difference quotient in (142) is exactly

$$\int_I \frac{q_{t+h}(v) - q_t(v)}{h} c_{t,t+h}(u,v)\,\mathrm{d}v \tag{143}$$

$$+ \frac{\int_I q_t(v) c_{t,t+h}(u,v)\,\mathrm{d}v - q_t(u)}{h}. \tag{144}$$

Lemma A.1 and (140) show that (143) tends to $\dot{q}_t(u)$. Definition (139) shows that (144) tends to $\mathcal{A}_t q_t(u)$. Hence

$$b_X(t, Q_t(u)) = \partial_t q(t,u) + \mathcal{A}_t q(t,\cdot)(u) = (\partial_t + \mathcal{A}_t) q(t,u). \tag{145}$$

Because U_t is uniform, the statement is naturally a Leb-almost-everywhere statement in u. A continuous kernel version extends it to every interior state where the limits exist. Strict monotonicity of q_t then transfers the sign of the right-hand side to the physical state $x = q_t(u)$, which proves the sign assertion in Proposition 2.22 after the limiting-mean condition is imposed separately. □

A.4 Conditional expectations and observable transition densities

Detailed proof of Theorems 2.7 and 2.21, and Proposition 2.8. For $s < t$, define

$$H_{s,t}(u) = \int_I q_t(v) c_{s,t}(u,v)\,\mathrm{d}v.$$

Column normalization and Tonelli's theorem give

$$\begin{aligned}
\int_I |H_{s,t}(u)|\,\mathrm{d}u &\le \int_I \int_I |q_t(v)| c_{s,t}(u,v)\,\mathrm{d}v\,\mathrm{d}u \\
&= \int_I |q_t(v)| \left\{ \int_I c_{s,t}(u,v)\,\mathrm{d}u \right\} \mathrm{d}v \\
&= \int_I |q_t(v)|\,\mathrm{d}v < \infty.
\end{aligned}$$

For a bounded $\mathcal{F}_s$-measurable random variable Z, the Markov property, first for bounded truncations of q_t and then by L^1 convergence, gives

$$\begin{aligned}
\mathbb{E}[Z X_t] &= \mathbb{E}[Z q_t(U_t)] \\
&= \mathbb{E}\left[Z \int_I q_t(v) c_{s,t}(U_s, v)\,\mathrm{d}v \right] = \mathbb{E}[Z H_{s,t}(U_s)].
\end{aligned}$$

This is the defining test-function identity for

$$\mathbb{E}[X_t \mid \mathcal{F}_s] = H_{s,t}(U_s).$$

Since $X_s = q_s(U_s)$ and $U_s \sim \text{Leb}$, equality $H_{s,t}(U_s) = q_s(U_s)$ almost surely is equivalent to $H_{s,t}(u) = q_s(u)$ for Leb-almost every u. This proves both directions of Theorem 2.7 for the fixed pair (s,t); no common null set over the uncountable collection of time pairs is asserted. Replacing the last equality by

$$H_{s,t}(u) - m = \mathcal{R}_{s,t}\{q_s(u) - m\}$$

proves both directions of Theorem 2.21.

Under the density assumptions of Proposition 2.8, the substitution $v = F_t(y),\ \mathrm{d}v = p_t(y)\,\mathrm{d}y$, in the conditional kernel gives

$$p^X_{s,t}(x,y) = c_{s,t}\{F_s(x), F_t(y)\}p_t(y).$$

The three required identities follow by explicit changes of variables:

$$\begin{aligned}
\int_{E_t^\circ} p^X_{s,t}(x,y)\,\mathrm{d}y &= \int_I c_{s,t}\{F_s(x), v\}\,\mathrm{d}v = 1,\\
\int_{E_s^\circ} p_s(x)p^X_{s,t}(x,y)\,\mathrm{d}x &= p_t(y)\int_I c_{s,t}\{u, F_t(y)\}\,\mathrm{d}u = p_t(y),\\
\int_{E_r^\circ} p^X_{s,r}(x,z)p^X_{r,t}(z,y)\,\mathrm{d}z &= p_t(y)\int_I c_{s,r}\{F_s(x), w\}c_{r,t}\{w, F_t(y)\}\,\mathrm{d}w\\
&= p_t(y)c_{s,t}\{F_s(x), F_t(y)\} = p^X_{s,t}(x,y).
\end{aligned}$$

The first line holds for every selected x, the second for Lebesgue-a.e. y, and the third for every selected x and Lebesgue-a.e. y. The final equality is precisely the density Chapman–Kolmogorov identity; integrating it over an arbitrary target Borel set yields the corresponding pointwise kernel identity. □

A.5 The generator martingale theorem and rescaling PDE

Detailed proof of Theorem 2.10. Write

$$B(t,u) = (\partial_t + \mathcal{A}_t)q(t,u).$$

Membership in the time–space extended generator domain means that, after a localizing sequence if necessary,

$$L_t = q(t,U_t) - q(0,U_0) - \int_0^t B(s,U_s)\,\mathrm{d}s \tag{146}$$

is a local martingale and the finite-variation integral is locally finite. If $B = 0$ for $\mathrm{d}t \otimes \mathrm{Leb}(\mathrm{d}u)$-almost every (t,u), uniformity of U_s and Tonelli's theorem yield, for every $T < \infty$,

$$\mathbb{E}\int_0^T \mathbf{1}_{\{B(s,U_s)\neq 0\}}\,\mathrm{d}s = \int_0^T\int_I \mathbf{1}_{\{B(s,u)\neq 0\}}\,\mathrm{d}u\,\mathrm{d}s = 0.$$

Thus the integral in (146) vanishes and $q(t,U_t)$ is a local martingale. Class–(D) on each finite horizon makes the stopped family uniformly integrable, so the local martingale is a true martingale.

Conversely, suppose $q(t,U_t)$ is a true martingale. It is also a local martingale, and subtracting the local martingale L in (146) shows that

$$A_t := \int_0^t B(s,U_s)\,\mathrm{d}s$$

is a local martingale. It is an absolutely continuous finite-variation process starting from zero. A finite-variation local martingale is indistinguishable from a constant, so $A_t = 0$ for all t almost surely and $B(s,U_s)=0$ for $\mathrm{d}s\otimes\mathbb{P}$-almost every (s,ω). Uniformity of U_s and Tonelli's theorem now give

$$0=\int_0^T \mathbb{P}\{B(s,U_s)\neq 0\}\,\mathrm{d}s=\int_0^T\int_I \mathbf{1}_{\{B(s,u)\neq 0\}}\,\mathrm{d}u\,\mathrm{d}s.$$

Hence $B=0$ for $\mathrm{d}t\otimes\mathrm{Leb}(\mathrm{d}u)$-almost every (t,u), proving the converse in Theorem 2.10. $\square$

Detailed generator proof of Proposition 2.20. For completeness, define the deterministic scaling factor

$$R_t=\mathcal{R}_{0,t}=\exp\left\{\int_0^t \gamma(a)\,\mathrm{d}a\right\},$$

and set

$$\widetilde{q}(t,u)=\frac{q(t,u)-m}{R_t}.$$

Since R_t is deterministic, $R_t'=\gamma(t)R_t$, and $\mathcal{A}_t$ acts only on the state variable,

$$(\partial_t+\mathcal{A}_t)\widetilde{q}(t,u)=\frac{1}{R_t}\left[(\partial_t+\mathcal{A}_t)q(t,u)-\gamma(t)\{q(t,u)-m\}\right]. \tag{147}$$

Thus the rescaled process is a martingale, subject to the preceding domain and class–(D) qualifications, exactly when (37) holds. At the finite-horizon level, multiplicativity gives $R_t=R_s\mathcal{R}_{s,t}$ and therefore

$$\mathbb{E}\left[\frac{X_t-m}{R_t}\middle|\mathcal{F}_s\right]=\frac{\mathcal{R}_{s,t}(X_s-m)}{R_s\mathcal{R}_{s,t}}=\frac{X_s-m}{R_s}.$$

The reverse calculation proves the converse part of Proposition 2.20 without appealing only to formal PDE algebra. $\square$

A.6 Diffusion transforms and reset mixtures

Detailed proof of Theorem 3.1. For Theorem 3.1, Itô's formula gives

$$\begin{aligned}\mathrm{d}\{f(t,R_t)-m\}&=(\partial_t+\mathcal{L})f(t,R_t)\,\mathrm{d}t+\sigma(R_t)f_x(t,R_t)\,\mathrm{d}W_t\\&=\gamma(t)\{f(t,R_t)-m\}\,\mathrm{d}t+\sigma(R_t)f_x(t,R_t)\,\mathrm{d}W_t.\end{aligned}$$

With $R_t^\gamma=\exp\{\int_0^t\gamma(a)\,\mathrm{d}a\}$, the deterministic product rule yields

$$\mathrm{d}\left[\frac{f(t,R_t)-m}{R_t^\gamma}\right]=\frac{\sigma(R_t)f_x(t,R_t)}{R_t^\gamma}\,\mathrm{d}W_t. \tag{148}$$

Condition (53) makes the stochastic integral in (148) square integrable on every finite horizon. Taking the conditional expectation between s and t and using $R_t^\gamma/R_s^\gamma=\mathcal{R}_{s,t}$ gives (54). Strict monotonicity gives the time-dependent inverse $R_s=f^{-1}(s,Y_s)$ and the transition kernel

$$P_{s,t}^Y(y,B)=P_{s,t}^R\big(f^{-1}(s,y),f^{-1}(t,B)\big),$$

which proves the Markov claim. Conversely, the conditional-mean identity makes the left side of (148) a martingale; uniqueness of the finite-variation term in Itô's decomposition forces (52) wherever the coefficients and derivatives are defined. $\square$

Detailed proof of Theorems 4.1 and 4.2. Finally, write $a=a_{s,r}$ and $b=a_{r,t}$. Invariance gives $P_{s,r}\Pi=\Pi P_{r,t}=\Pi$ and $\Pi^2=\Pi$, so the reset-kernel composition in Theorem 4.1 expands as

$$\begin{aligned}&\{aP_{s,r}+(1-a)\Pi\}\{bP_{r,t}+(1-b)\Pi\}\\&\quad=abP_{s,t}+\{a(1-b)+(1-a)b+(1-a)(1-b)\}\Pi\\&\quad=abP_{s,t}+(1-ab)\Pi=P_{s,t}^{\mathrm{mix}},\end{aligned}$$

because $ab = a_{s,t}$. For the Copula densities the corresponding integral is

$$
\begin{aligned}
&\int_I \{ac_{s,r}(u,w) + 1 - a\}\{bc_{r,t}(w,v) + 1 - b\}\,\mathrm{d}w \\
&\quad = abc_{s,t}(u,v) + a(1-b) + (1-a)b + (1-a)(1-b) \\
&\quad = a_{s,t}c_{s,t}(u,v) + 1 - a_{s,t},
\end{aligned}
$$

where row normalization, column normalization, and the base Chapman–Kolmogorov identity were used term by term. This density equality is for every selected u and a.e. v; its integrated kernel equality is for every u and every Borel target set. Moreover, $a_{t,t+h} = 1 - \lambda(t)h + o(h)$ and $P_{t,t+h}h_0 = h_0 + h\mathcal{G}_t h_0 + o(h)$ on the generator domain. Therefore

$$
\frac{P^{\mathrm{mix}}_{t,t+h}h_0 - h_0}{h} = \mathcal{G}_t h_0 + \lambda(t)\{\Pi h_0 - h_0\} + o(1),
$$

which proves (103). The conditional centered first moment is, directly in integral form,

$$
\begin{aligned}
\int_E (y-m)P^{\mathrm{mix}}_{s,t}(x,\,\mathrm{d}y) &= a_{s,t}\int_E (y-m)P_{s,t}(x,\,\mathrm{d}y) \\
&\quad + (1-a_{s,t})\int_E (y-m)\pi(\,\mathrm{d}y) \\
&= a_{s,t}\mathcal{R}_{s,t}(x-m),
\end{aligned}
$$

and its right derivative at $t = s$ gives $b_{\mathrm{mix}}(s,x) = b_0(s,x) + \lambda(s)(m-x)$. $\square$

B Theory and implementation audit checklist

✓ MOU density equals $e^{-\lambda h} p_{\mathrm{OU}} + (1 - e^{-\lambda h}) f_{\mathrm{OU,stat}}$ pointwise. The code test verifies exact equality to OU when $\lambda = 0$.
✓ MCIR uses the stationary Gamma law implied by the same (θ, κ, σ) as the CIR base.
✓ Mixed conditional mean decays at $\kappa + \lambda$; simulation and analytic moments are compared in the test suite and generator-validation table.
✓ Affine OU and CIR inverse transformations and Jacobians match the article's linear formulas under $A_0 = 1$ and $B_0 = 0$.
✓ All primary five-minute pairs are exact timestamp matches; no missing bar is treated as an observed transition.
△ The article assumes a stationary base process. Rolling estimates show substantial parameter variation, so stationarity is an empirical approximation rather than an established fact.
△ The semiparametric empirical marginal does not impose the article's Gaussian-copula marginal sign restriction and fails the finite-horizon sign diagnostic on most grid points.
△ Price-CIR likelihood evaluation requires a high-noncentrality Gaussian moment fallback for all evaluated price transitions; price CIR and affine transformed CIR are flagged as weakly identified.
✓ The primary boundary bootstrap uses the same 25,000 transition pairs and the same joint OUF/MOUF estimators for the observed and all 500 simulated statistics. Its inference is conditional on the systematic design and fixed observed regressors, rather than an unconditional full-sample statement.
△ The two-scale scalar copula model captures multi-horizon dependence but is not one-dimensional Markov; it is a diagnostic latent-factor extension rather than an implementation of the article's scalar Markov theorem.
✓ Deterministic seasonal $c(t)$, exponential OU, quadratic and exponential CIR, and the article's special-function branches are implemented. Converged, weak, non-monotone, and support-failed outcomes are all retained in the same implementation-status table.
✓ Normal, Student-t, skew Student-t, GED, and monotone-spline margins are crossed with Gaussian, Student-t, Frank, Clayton, survival Clayton, Gumbel, and survival Gumbel copulas; base and independence-mixed versions are scored for all three periods and four horizons.
✓ Random-walk, jump-OU, nested-OU, two-state OU, and a finite-dimensional deterministic time-varying marginal–copula model enter the same primary validation ranking. No model is removed because it is complex.
△ Markov-switching OU wins the exact filtered one-step score but its slow-regime rate is near the optimization boundary and spans substantially less than one mean-reversion unit in the continuous estimation segment.
× The mixed-copula bond-pricing PDE is not empirically observable from a USDT spot-price series without a bond-price cross section and a specified risk-neutral measure. This is retained as an explicit data-identification failure rather than an unimplemented model.

C Output provenance

Table 14: Primary reproducibility files

File	Content
`output/tables/data_audit.csv`	Raw-data hash, rows, timestamps, gaps, and audit conditions
`output/tables/sample_summary.csv`	Sample periods, coverage, and exact-pair counts
`output/tables/sign_test_baseline.csv`	Nonparametric sign-consistency statistics

File	Content
output/tables/model_estimates.csv	Parameters and ordinary, cluster, and bootstrap uncertainty
output/tables/model_comparison.csv	Likelihood, information criteria, numerical checks, and PIT tests
output/tables/matched_joint_mou_lr_bootstrap.csv	Primary matched-design joint-MOUF boundary bootstrap
output/tables/matched_joint_mou_lr_bootstrap_draws.csv	All 500 null LR draws and convergence diagnostics
output/tables/mixed_lr_bootstrap.csv	Superseded restricted boundary-bootstrap sensitivity
output/tables/copula_horizon_fit.csv	Empirical and fitted dependence over six horizons
output/tables/copula_scale_model_comparison.csv	Single/two-scale fit, parameters, and dependence RMSE
output/tables/copula_scale_horizon_fit.csv	Four-model fitted dependence at each horizon
output/tables/copula_scale_oos_scores.csv	Frozen-training-model validation and test copula log scores
output/tables/article_model_hierarchy.csv	Structural, extension, diagnostic, benchmark, and appendix tiers
output/tables/copula_conditional_drift_baseline.csv	Semiparametric conditional means and sign diagnostic
output/tables/forecast_scores.csv	Validation and test forecast scores
output/tables/first_passage_comparison.csv	Empirical and simulated recovery-time distributions
output/tables/robustness_estimates.csv	Frequency, extended-sample, and anomaly-policy robustness
output/tables/full_model_registry.csv	Frozen article/methodology model universe and comparison groups
output/tables/full_model_implementation_status.csv	Convergence, weak identification, monotonicity, support, and retained failures
output/tables/full_model_validation_ranking.csv	Complexity-blind 2024 five-minute conditional-density ranking
output/tables/full_model_pairwise_validation.csv	Paired day-block uncertainty for the selected model
output/tables/full_mixture_incremental_scores.csv	Within-family mixed-minus-base score increments
output/tables/updated_stage_gate.csv	Full-model coverage, scoring, diagnostics, and no-leakage gates
output/models/mixed_fits_baseline.json	Full MOU/MCIR parameter and optimization records
output/models/full_model_fits.json	Full-suite fits and nonlinear failure diagnostics
output/models/full_copula_fits.json	Five margins and fourteen fitted copula specifications
output/models/semiparametric_copula_baseline.json	Multi-horizon composite-likelihood estimate
output/models/copula_scale_models.json	Full-sample and training-sample scale/reset estimates
output/run_status.json	Step-level one-click pipeline status
output/manifest.json	Artifact hashes and joint test/pipeline completion status

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